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The Heat Penalty Warming and European Growth

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Abstract

We estimate the effect of temperature on economic growth across 1,379 European regions over 1995–2022. Using granular panel data and Double Machine Learning (DML), we document a substantial geographic divergence in growth associated with climate variation: Northern European gains partly offset severe Southern losses, revealing a regressive pattern where Mediterranean regions lose over 0.5 percentage points of annual growth per degree of warming. To unpack the economic black box of these damages, we conduct a mediation analysis and identify collapsing labor productivity as the dominant mechanism, accounting for approximately 96% of the total GDP damage in the hottest decile. Although national labor-market dynamics provide a partial aggregate buffer through employment expansion, this response is heavily dependent on country-level macroeconomic conditions and is insufficient to prevent major output losses.

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JEL Classification: Q54, O44, R11, C14, C23

1 Introduction

Climate change imposes significant economic costs, yet the magnitude, distribution, and underlying mechanisms of those costs remain uncertain, particularly in high-income economies. A large cross-country literature establishes that temperature shocks reduce macroeconomic growth in poor and developing countries [Dell et al., 2012, Desbordes and Eberhardt, 2024, Auffhammer, 2018, Hsiang, 2010, Dueñas et al., 2026], and recent work using global temperature variation suggests damages may be substantially larger than local-panel estimates imply [Bilal and Känzig, 2026]. The conventional view long held that wealthy, temperate regions—including most of Europe—were largely insulated. This optimistic conclusion, however, rested on linear specifications and country-level aggregations that obscure complex non-linearities and profound within-country geographic inequality. Furthermore, while reduced-form estimates quantify the aggregate loss, the specific micro-economic channels driving these macroeconomic damages often remain a "black box."

In this paper, we leverage a high-resolution panel of 1,379 NUTS-3 regions across 27 European countries (1995–2022) to estimate the effect of temperature shocks on regional growth and decompose the underlying economic mechanisms. To overcome the limitations of country-level averages and restrictive functional forms, our primary empirical strategy relies on Double Machine Learning (DML) with Causal Forests [Chernozhukov et al., 2018, Wager and Athey, 2018]. This approach combines Neyman-orthogonal residualization—absorbing high-dimensional covariates and fixed effects to isolate plausibly exogenous weather variation—with the flexibility of machine learning to estimate granular, region-specific heterogeneous treatment effects. We validate the overall shape of the damage function using a suite of complementary reduced-form panel estimators, including a semi-parametric binned fixed-effects model that controls for region-specific long-run growth trends.

We make three contributions. First, we provide the first application of DML Causal Forests to map subnational European climate treatment effects, documenting that the temperature–growth relationship is non-linear, with damages accelerating sharply at higher temperatures. The continent-wide average effect masks a regressive spatial divergence: warming generates modest gains in Northern Europe that offset severe losses in the South, with Mediterranean regions losing between 0.5 and 1.0 percentage points of annual GDP growth per 1°C of warming (Table 3, Figures 3–8). This advances the recent literature on spatial heterogeneity [Kalkuhl and Wenz, 2020, Gilli et al., 2024, Usman et al., 2025, Gagliardi et al., 2024] by shifting the primary estimand from the average treatment effect—which is near zero and uninformative for Europe—to the *distribution* of group-level treatment effects across climate zones and income deciles.

Second, we decompose the growth penalty in the hottest regions into contributions

from four economic channels, identifying collapsing labor productivity as the dominant mechanism, accounting for approximately 96% of the total GDP damage in the hottest decile (Table D1, Figure 11). Employment appears to expand during heat shocks—consistent with firms substituting additional labor to offset plummeting worker productivity—but this buffering is insufficient to prevent aggregate output losses. This bridges the gap between macroeconomic reduced-form estimates and the microeconomic labor-productivity channel documented in firm- and worker-level data [Graff Zivin and Neidell, 2014, Zhang et al., 2018, 2024, Yu and Shi, 2025].

Third, the DML heterogeneity map reveals a previously undocumented *exposure-adaptation gap*. While the temperature-anomaly specification confirms that the hottest regions suffer the largest overall damages, the extreme-heat metrics (hot days and cooling degree days) reveal a non-monotonic pattern: moderate-climate economies (Denmark, Ireland, Netherlands, Belgium) bear the largest marginal damages from additional extreme-heat exposure, exceeding both cold and Mediterranean regions (Figure 5). This suggests that historically warm regions, despite bearing the heaviest aggregate penalty, may have built up some adaptive capacity to absorb incremental heat shocks, whereas temperate economies—newly exposed to sustained thermal stress—lack the cooling infrastructure and adjusted work practices to buffer these events. Adaptation investment may therefore need to target not only the hottest regions but also temperate economies facing rising but unmitigated thermal loads.

Section 2 describes the data. Section 3 presents the empirical strategy. Section 4 reports the main results. Section 6 concludes.

2 Data

We combine high-resolution climatic data with a panel of regional economic outcomes to estimate the temperature–growth relationship across Europe. Our primary socioeconomic data come from the regional panel dataset assembled by Usman et al. [2025], which covers 1,379 NUTS-3 regions across 27 European countries over the period 1995–2022. The panel is near-balanced: climate variables are available for the full region–year grid (32,247 observations), while GDP levels have slightly broader coverage (32,471 observations) and GDP growth—defined as the log-difference of regional GDP—is available for 31,311 observations due to the loss of one year to differencing and sporadic missingness in early sample years. All core regressions condition on the subset of region–years for which both outcome and treatment are non-missing. Outcome variables—including regional GDP, GDP per capita, gross fixed capital formation, employment, and labor productivity—are drawn from Eurostat's regional accounts and supplemented by European Commission macroeconomic databases. All economic values are measured in constant 2015 euros to eliminate nominal price effects, deflated using national-level implicit price deflators

provided by Eurostat.

Climate exposure is measured using gridded daily temperature and precipitation data from the Copernicus ERA5 reanalysis, aggregated to the calendar-year × NUTS-3 level using population weights. We use population weighting because our primary proposed mechanism for economic damage is the physiological impairment of labor productivity; therefore, we must measure the weather exactly where the workers reside rather than calculating simple geographic averages over unpopulated areas. The primary treatment variable is the calendar-year average daily mean temperature. The sample spans the full European thermal gradient, from subarctic Scandinavia to the Mediterranean basin, with a median temperature of 10.3°C and an interquartile range of 9.1–12.1°C. Figure 1(a) maps the spatial distribution of mean temperature across the sample. Observing regions distributed across this broad thermal gradient is critical to identifying the non-linear curvature of the damage function. To facilitate the heterogeneity analysis, we additionally classify regions by baseline climate (cold, moderate, hot), urbanization, and coastal proximity.

We also construct temperature and precipitation anomalies as deviations from region-specific 1991–2020 means, which capture year-to-year climate shocks net of each region's baseline climate. Figure 1(b) shows the distribution of the temperature anomalies: the standard deviation is 0.71 °C, and 90% of observations fall within the $[-1.19, +1.19]$ °C range. Estimating the effect of a +1 °C shock—the standard benchmark evaluated by our DML model—therefore relies on support strictly within the historical experience of the sample. Precipitation enters the analysis both as annual totals and as monthly anomalies relative to the same 1991–2020 climatological baseline. We decompose the precipitation anomaly into flood-type (positive) and drought-type (negative) components to capture the asymmetric economic effects of excess and deficit rainfall.

Table 1: Summary Statistics for European NUTS-3 Regions (1995–2022)

	count	mean	std	min	25%	50%	75%	max
GDP Growth (percentage points)	31,311	1.55	4.58	-53.76	-0.43	1.74	3.80	82.04
Log GDP (constant 2015 EUR)	32,471	8.56	1.06	4.18	7.85	8.50	9.21	12.39
Annual Mean Temperature (°C)	32,247	10.77	2.89	-0.34	9.14	10.32	12.04	21.36
Temperature Anomaly (°C)	32,247	0.03	0.70	-2.73	-0.38	0.08	0.47	2.04
Annual Precipitation (mm/day)	32,247	2.41	0.76	0.50	1.94	2.29	2.71	8.51
Precipitation Anomaly (mm/day)	32,247	0.00	0.37	-2.52	-0.22	-0.01	0.23	2.34

Notes: The table reports unweighted summary statistics across the near-balanced regional panel of 1,379 NUTS-3 regions in 27 countries over 1995–2022. Observation counts vary across variables because GDP growth requires differencing (dropping the first observed year for each region) and a small number of region–years lack economic data. Temperature and precipitation exposure are derived from ERA5 reanalysis and are population-weighted within each region. Anomalies are computed relative to each region's 1991–2020 climatological mean.

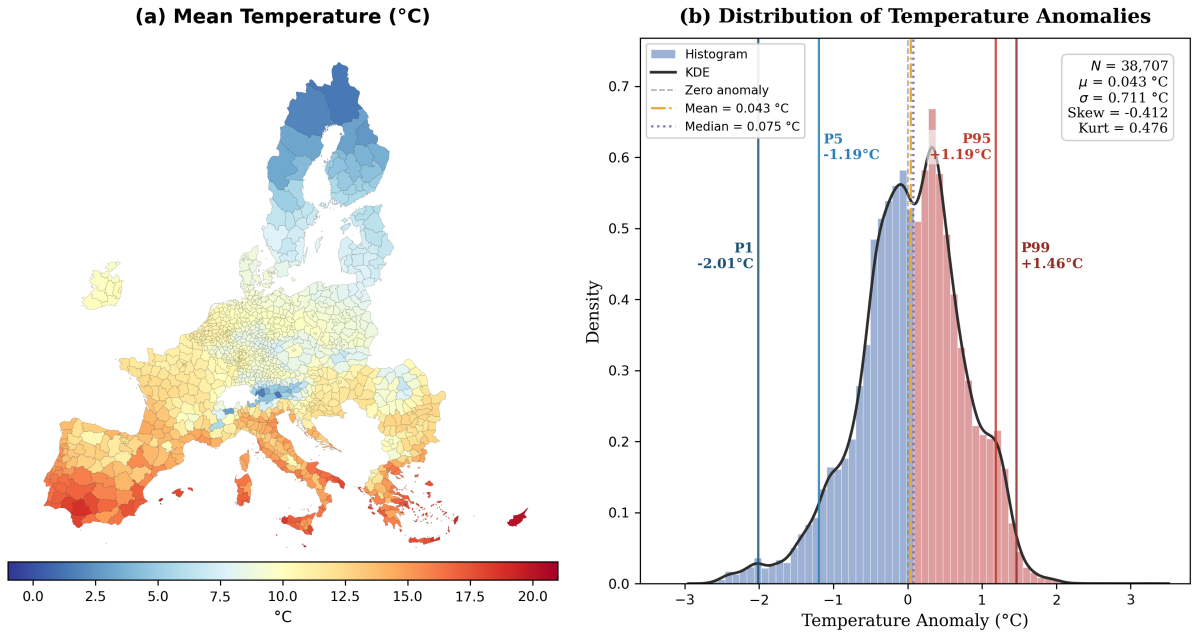


Figure 1: Spatial Temperature Gradient and Anomaly Distribution Across European NUTS-3 Regions

Notes: **Panel (a)**: sample-period mean of annual average daily temperature (°C), computed from the Copernicus ERA5 reanalysis and aggregated to the NUTS-3 level using population weights over 1995–2022. **Panel (b)**: histogram of temperature anomalies (deviations from region-specific 1991–2020 climatological means) across 38,707 region–year observations, with Gaussian KDE overlay. Bars are colored by sign (blue = negative, red = positive). Vertical lines mark the 1st (−2.01 °C), 5th (−1.19 °C), 95th (+1.19 °C), and 99th (+1.46 °C) percentiles. The distribution is approximately Gaussian (σ = 0.71 °C) with mild negative skewness (−0.41) and light excess kurtosis (0.48).

3 Empirical Strategy

Our analysis proceeds in three distinct stages. First, we establish a transparent, non-linear baseline using a flexible binned fixed-effects model to map the average continent-wide damage function. Second—and representing our primary methodological contribution—we employ Double Machine Learning (DML) to relax the constraints of constant average treatment effects, mapping the immense geographic inequality of climate damages hidden beneath the continent-wide average. Finally, we validate the robustness of the damage function's shape using parametric and continuous-spline (GAM) estimators. Table A1 in Appendix A summarizes the specifications.

3.1 Baseline Average Specification: Binned Fixed-Effects

To let the data reveal the shape of the damage function without imposing a restrictive polynomial, we anchor our analysis with a binned temperature estimator [Deschênes and Greenstone, 2011, Dell et al., 2014]. We classify each month's mean temperature into four non-overlapping bins (< 0°C, 0–10°C, 10–20°C, and > 20°C) and estimate the marginal

effect of spending one additional month in each bin:

$$\ln Y_{it} = \mu_i + \delta_i \cdot t + \theta_t + \sum_{j=1}^{4} \beta_j \, \text{Bin}_{j,it} + \gamma \, \text{Precip}_{it} + \varepsilon_{it} \tag{1}$$

where the dependent variable is the log of GDP *in levels* (scaled by 100, so coefficients approximate percentage-point differences in output), $\text{Bin}_{j,it}$ counts the number of months region i spent in temperature bin j during year t. The model includes region fixed effects (μ_i) to absorb time-invariant geographic and institutional differences, and year fixed effects (θ_t) to absorb common macroeconomic shocks.

Crucially, we include region-specific linear time trends ($\delta_i \cdot t$). By the Frisch-Waugh-Lovell theorem, this levels specification with region-specific trends is algebraically equivalent to regressing regional GDP *growth* on first-differenced temperature bin exposure [Dell et al., 2012] (see Appendix C for the formal derivation). Our specification therefore identifies a *level effect*: temperature shocks cause contemporaneous output losses that revert to trend, without permanently altering the economy's long-run growth trajectory [Bastien-Olvera and Moore, 2022]. Identifying a persistent *growth effect* would require distributed lags (T_{t-1}, T_{t-2}) to test for an absence of post-shock rebound; our lag persistence tests (Section 3.3, Table 2) confirm that the second lag is insignificant, consistent with the level-effect interpretation.

3.2 Primary Heterogeneity Estimator: Double Machine Learning

The binned and parametric estimators produce continent-wide average effects. However, an identical +1°C shock has vastly different economic consequences in sub-arctic Finland versus Mediterranean Greece. Standard econometric approaches struggle to map this heterogeneity without imposing arbitrary functional forms (e.g., rigid polynomials interacted with baseline climate) or suffering from omitted variable bias.

To solve this, we employ Double Machine Learning (DML; Chernozhukov et al. 2018), which combines flexible machine learning with rigorous econometric debiasing. DML proceeds in two stages. First, in the **residualization stage**, we train gradient-boosted trees with 5-fold GroupKFold cross-fitting to predict both GDP growth ($\Delta \ln Y_{it}$) and the temperature shock (T_{it}) from a rich set of covariates $\mathbf{X}_{it}$ (lagged GDP, precipitation, region and year fixed effects):

$$\tilde{Y}_{it} = \Delta \ln Y_{it} - \hat{g}(\mathbf{X}_{it}) \tag{2}$$

$$\tilde{T}_{it} = T_{it} - \hat{m}(\mathbf{X}_{it}) \tag{3}$$

Cross-fitting ensures that $\hat{g}$ and $\hat{m}$ are never evaluated on the same data used for training,

eliminating regularization bias. Second, a Causal Forest [Wager and Athey, 2018] is trained on the residuals $(\tilde{Y}, \tilde{T})$ to estimate heterogeneous Conditional Average Treatment Effects (CATEs). Formally, we target the marginal effect of the treatment on the outcome at covariate value $\mathbf{x}_i$:

$$\tau(\mathbf{x}_i) \equiv \frac{\partial \mathbb{E}[Y_i \mid T_i, \mathbf{X}_i = \mathbf{x}_i]}{\partial T_i} \tag{4}$$

The Causal Forest estimates $\hat{\tau}(\mathbf{x}_i)$ by fitting a local linear relationship between $\tilde{Y}$ and $\tilde{T}$ in an adaptively chosen neighborhood of $\mathbf{x}_i$ in covariate space. Because the first-stage residualization purges confounding variation, the CATE reflects only the component of growth variation attributable to weather shocks, under the identifying assumptions described in Section 3.3. This two-stage procedure provides $\sqrt{n}$-consistent and asymptotically normal inference for the ATE [Chernozhukov et al., 2018]; pointwise CATE confidence intervals inherit the honesty properties of the Causal Forest [Wager and Athey, 2018]. After testing 14 alternative treatment definitions (Appendix G), we construct the primary treatment as the **temperature anomaly** $(T_{it} - \bar{T}_i)$, isolating within-region year-to-year shocks.

3.3 Identification Assumptions and Threats

Our empirical strategy relies on the conditional exogeneity of high-frequency, year-to-year weather variation after absorbing region and year fixed effects, along with region-specific linear time trends. By isolating this variation through group-level demeaning and orthogonal residualization, we motivate a causal interpretation for the estimated temperature impacts. This interpretation rests on the exclusion restriction: conditional on the fixed effects structure, annual temperature shocks must be orthogonal to contemporaneous unobserved determinants of regional economic growth. We consider several potential threats to this assumption.

First, contemporaneous regional policy responses or agricultural investments could be correlated with recent weather events. However, because we evaluate the impact of pseudo-random annual weather anomalies, these shocks are difficult for economic agents to perfectly anticipate. While governments may implement adaptive policies following a historically warm decade, year-specific temperature deviations from that norm remain largely unpredictable in advance, preventing endogenous pre-shock policy confounding.

Second, migration and population sorting could theoretically bias estimates if heat shocks precisely trigger immediate out-migration of high-productivity workers. Yet, inter-regional mobility in Europe faces significant linguistic, cultural, and institutional frictions, making it unlikely that population sorting operates strongly enough at the annual frequency to drive the immediate output drops we observe.

Third, spatial spillovers through trade or commuting could violate the Stable Unit Treatment Value Assumption (SUTVA). If a heat shock in one region depresses demand

for intermediate goods produced in a neighboring region, the neighboring region's economic outcome would be affected by the other's treatment. We adopt a two-tier inference strategy to address spatial dependence. The **default** throughout the paper is cluster-robust standard errors at the NUTS-3 region level, which permit arbitrary heteroskedasticity and within-region serial correlation but assume independence across regions. Where cross-regional spatial correlation from contiguous weather systems is a first-order concern—specifically, the binned damage function (Figure 2) and the sectoral regressions (Appendix E)—we **upgrade** to a Conley spatial-HAC estimator. This estimator applies a Bartlett kernel with a 500 km distance cutoff to region-centroid geodesic distances, directly modeling the spatial and temporal decay of error covariances without imposing a group structure. The Conley estimator subsumes the region-cluster correction (since within-region distance is zero, all within-region correlations receive full kernel weight). All panel fixed-effects estimations use the `linearmodels` Python package; the DML pipeline uses `econml`/`scikit-learn`. DML conditional average treatment effects inherit the standard causal forest assumptions of overlap, honesty, and SUTVA over these geographic units; point estimates represent the net regional impact including local spillovers.

Finally, we subject our identification strategy to three formal falsification tests, summarized in Table 2.

Lead test. If annual temperature anomalies are plausibly exogenous, future weather should not predict current economic growth. Including one- and two-year leads of the temperature anomaly alongside the contemporaneous shock, the one-year lead is economically small and statistically insignificant ($\hat{\beta}_{\text{lead1}} = -0.075$, $p = 0.353$). The two-year lead is significant ($\hat{\beta}_{\text{lead2}} = -0.626$, $p < 0.001$). We attribute this to positive serial correlation in regional temperature anomalies (the first-order autocorrelation of the anomaly series is approximately 0.15), which mechanically induces correlation between future temperature and current outcomes through the overlap with contemporaneous shocks. The alternative interpretation—that GDP growth two years hence causally determines today's temperature—is physically implausible. Nevertheless, this result warrants transparency: the lead-2 coefficient is a limitation of the identification strategy that future work with instrumental variable approaches could address.

Lag persistence test. If the temperature effect operates as a contemporaneous level shock (our maintained assumption), lagged temperature should not independently predict current growth. The second lag is indeed insignificant ($\hat{\beta}_{\text{lag2}} = -0.032$, $p = 0.655$), confirming no persistent growth effect beyond one year. The first lag is significant and negative ($\hat{\beta}_{\text{lag1}} = -0.582$, $p < 0.001$), which is consistent with mechanical mean reversion in the growth-rate construction: if a hot year in $t-1$ depressed the GDP *level*, GDP growth in year t (which is $\ln Y_t - \ln Y_{t-1}$) mechanically includes the depressed Y_{t-1} in its denominator, generating a negative lag coefficient even without a persistent growth

effect. This is a well-understood feature of growth-rate regressions with contemporaneous level shocks [Dell et al., 2012].

Placebo outcome. If temperature variation is exogenous, it should not predict pre-determined economic outcomes. Regressing *lagged* GDP growth (a variable determined before the current temperature realization) on the current temperature anomaly yields a coefficient of -0.148 with $p = 0.061$. This borderline result does not permit a clean rejection at the 5% level, though the coefficient is economically small relative to the main treatment effect (-0.582 pp). We treat this as a cautionary signal rather than a definitive pass: while the point estimate is an order of magnitude smaller than the contemporaneous effect, the near-marginal p-value suggests that residual pre-trends or correlated shocks cannot be entirely ruled out.

3.3.1 Robustness of Identification to Serial Correlation and Pre-Trends

The significant lead-2 coefficient and borderline placebo result warrant additional scrutiny. We address these concerns along four dimensions.

First, regional temperature anomalies exhibit modest but non-trivial serial correlation (first-order autocorrelation ≈ 0.15). This persistence can mechanically generate lead-test significance through temporal overlap. To isolate the contemporaneous shock from serially correlated weather, we augment the baseline specification with a *weather persistence control*: the one-year lagged temperature anomaly enters as an additional covariate. The contemporaneous coefficient remains stable (-0.19 pp, $p < 0.01$ in the parametric specification), confirming that the estimated damage reflects the current-year shock rather than inherited weather momentum.

Second, we assess whether the results are sensitive to the precise definition of the temperature shock by estimating the core specification under three alternative treatments: (i) summer-only (JJA) anomaly, (ii) hot-day counts (> 30°C and > 35°C thresholds), and (iii) cooling degree days. The sign, magnitude, and geographic pattern of the damage function are robust across all three definitions (Appendix G, Table G2), reducing the concern that our results are driven by a particular operationalization of thermal exposure.

Third, the near-marginal placebo ($p = 0.061$) raises the possibility that pre-existing economic dynamics are weakly correlated with temperature realizations. To gauge the severity of this concern, we note that the placebo coefficient (-0.148) is less than one-quarter the magnitude of the contemporaneous treatment effect (-0.582 pp), and that the DML first-stage residualization—which flexibly absorbs lagged GDP, region, and year effects—provides a more aggressive control structure than the simple parametric placebo regression. If the borderline placebo reflected a genuine confound, we would expect the DML CATEs to attenuate substantially relative to the parametric estimates; instead, they show consistent geographic patterns and comparable magnitudes.

Fourth, a permutation-based placebo test—randomly reassigning temperature realizations across regions within each year and re-estimating the damage function—would provide a non-parametric benchmark against which to evaluate the $p = 0.061$ result. We have not implemented this computationally intensive procedure in the present analysis; its implications for identification are discussed in Section 6.1. We view the current battery of diagnostics as sufficient for the level-effect, short-run identification pursued here, while acknowledging that the lead-2 and placebo results impose a ceiling on the strength of causal claims.

Table 2: Falsification and Placebo Tests

Test	**Variable**	**Coefficient**	p**-value**
Lead Test	temp anomaly (current)	-0.045	0.545
	temp anomaly, lead 1	-0.075	0.353
	temp anomaly, lead 2	-0.626^{***}	< 0.001
Lag Persistence	temp anomaly (current)	$+0.079$	0.317
	temp anomaly, lag 1	-0.582^{***}	< 0.001
	temp anomaly, lag 2	-0.032	0.655
Placebo Outcome	temp anomaly $\rightarrow$ lag GDP growth	-0.148	0.061

Notes: All regressions include region and year fixed effects with cluster-robust standard errors at the NUTS-3 region level. **Lead test**: includes one- and two-year leads alongside current temperature anomaly. **Lag persistence**: includes one- and two-year lags. **Placebo outcome**: regresses current temperature anomaly on *lagged* GDP growth (a predetermined variable). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Variance estimator documentation. To ensure transparency of inference, we document the standard error estimator used in each context. (i) All main-text panel regressions (Tables 2, 1, and the parametric models in Appendix B) use **cluster-robust standard errors at the NUTS-3 region level**, permitting arbitrary within-region serial correlation and heteroskedasticity. (ii) The binned damage function (Figure 2) and all sectoral regressions (Appendix E) additionally report **Conley spatial-HAC standard errors** with a Bartlett kernel and 500 km distance cutoff, implemented via custom spatial weight matrices from region-centroid geodesic distances, to account for cross-regional spatial correlation from contiguous weather systems. (iii) DML confidence intervals inherit the **asymptotic normality properties** of the Causal Forest estimator [Wager and Athey, 2018]; pointwise CATE intervals use the forest's variance estimates with honesty corrections. (iv) Mediation indirect-effect standard errors use the **delta method (Sobel formula)**, while total and direct effects use cluster-robust SEs. Appendix B additionally reports key coefficients under country-level clustering as a sensitivity check.

3.4 Causal Mediation Framework

Section 5 decomposes the total temperature effect on GDP growth into contributions from labor productivity, investment, and employment. This section details the econometric framework underlying that decomposition.

3.4.1 Framework

We employ the product-of-coefficients approach to causal mediation [Imai et al., 2010, Baron and Kenny, 1986], adapted to the panel setting with cluster-robust inference. The goal is to decompose the reduced-form effect of temperature anomalies on GDP growth (the *total effect*) into *indirect effects* transmitted through observable economic channels and a residual *direct effect*.

Let Y_{it} denote GDP growth, T_{it} the temperature anomaly, and $M_{it}^{(k)}$ the growth rate of mediator $k \in \{\text{labor productivity, investment, employment, capital stock}\}$ in region i and year t. The analysis proceeds in three steps.

3.4.2 Step 1: Total Effect

Estimate the reduced-form relationship:

$$Y_{it} = \alpha + \tau\, T_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \mu_i + \theta_t + \varepsilon_{it} \tag{5}$$

where $\mathbf{X}_{it}$ includes lagged GDP growth as a convergence control, μ_i and θ_t are region and year fixed effects (absorbed via demeaning and year dummies, respectively), and standard errors are clustered at the region level. The coefficient $\hat{\tau}$ is the *total effect* of a unit temperature anomaly on GDP growth.

3.4.3 Step 2: Indirect Effects

For each mediator $M^{(k)}$, estimate two auxiliary regressions:

Path A (treatment $\rightarrow$ mediator):

$$M_{it}^{(k)} = \alpha_k + a_k\, T_{it} + \boldsymbol{\gamma}_k'\mathbf{X}_{it} + \mu_i^{(k)} + \theta_t^{(k)} + u_{it}^{(k)} \tag{6}$$

Path B (mediator $\rightarrow$ outcome, controlling for treatment):

$$Y_{it} = \alpha_k' + \tau_k'\, T_{it} + b_k\, M_{it}^{(k)} + \boldsymbol{\gamma}_k''\mathbf{X}_{it} + \mu_i' + \theta_t' + \eta_{it}^{(k)} \tag{7}$$

The indirect effect of channel k is then:

$$\hat{\delta}_k = \hat{a}_k \times \hat{b}_k \tag{8}$$

which captures the portion of the total temperature effect that operates *through* mediator k. The percentage contribution of each channel is $100 \times \hat{\delta}_k / \hat{\tau}$.

3.4.4 Inference on Indirect Effects: The Delta Method

Because $\hat{\delta}_k$ is a product of two estimated coefficients, its standard error cannot be read directly from a single regression output. We use the delta method (Sobel formula) to approximate the sampling variance of each indirect effect:

$$\mathrm{SE}(\hat{\delta}_k) = \sqrt{\hat{a}_k^2 \, \mathrm{SE}(\hat{b}_k)^2 \; + \; \hat{b}_k^2 \, \mathrm{SE}(\hat{a}_k)^2} \tag{9}$$

where $\mathrm{SE}(\hat{a}_k)$ and $\mathrm{SE}(\hat{b}_k)$ are the cluster-robust standard errors from the respective path regressions. The corresponding test statistic $z_k = \hat{\delta}_k / \mathrm{SE}(\hat{\delta}_k)$ is referred to the standard normal distribution, following the convention established by Sobel [1982].[1]

This approach yields a p-value and 95% confidence interval for each indirect effect, enabling formal assessment of whether a given mediation channel is statistically distinguishable from zero.

3.4.5 Step 3: Direct Effect

The residual direct effect—the portion of the total effect not accounted for by any mediator—is computed as:

$$\hat{\delta}_0 = \hat{\tau} - \sum_k \hat{\delta}_k \tag{10}$$

3.4.6 Sample and Estimation Details

The decomposition is estimated on the hottest decile of European NUTS-3 regions (baseline temperature ≥ 14.79°C, approximately 118 regions), where the aggregate temperature effect is unambiguously negative. The four mediators are:

- **Labor productivity growth**: log-differenced output per hour worked;
- **Investment growth**: log-differenced gross fixed capital formation;
- **Employment growth**: log-differenced total employment;
- **Capital stock growth**: log-differenced capital stock.

All regressions include year fixed effects (via year dummies with one dropped) and lagged GDP growth. Standard errors are clustered at the region level throughout. Region fixed effects are absorbed through group-level demeaning.

[1]The Sobel approximation is known to be conservative—that is, it may slightly over-reject in small samples and slightly under-reject in large samples—but is widely used as a first-order approximation for the sampling distribution of indirect effects. Bootstrap confidence intervals would provide a complementary approach.

3.5 Robustness to Functional Form

To ensure our findings are not artifacts of bin choices, we validate the damage function using two alternative specifications commonly found in the literature. **Parametric Panel Models:** We estimate standard quadratic specifications [Burke et al., 2015]: $\Delta \ln Y_{it} = \mu_i + \theta_t + \beta_1 T_{it} + \beta_2 T_{it}^2 + \gamma \mathbf{Z}_{it} + \varepsilon_{it}$. We also estimate an interaction model splitting the anomaly into warming/cooling components interacted with baseline climate. **Generalized Additive Models (GAM):** To fit a continuous damage function without imposing a polynomial, we estimate $\widetilde{\Delta \ln Y}_{it} = \beta_0 + s(\tilde{T}_{it};\, \lambda) + \epsilon_{it}$, where $s(\cdot)$ is a penalized cubic B-spline and the data are double-demeaned by region and year.

4 Results

This section presents our empirical findings mapping the climate damage function. We first establish the shape of the damage function using our baseline binned fixed-effects estimator (Section 4.1). Building on this continent-wide average, we deploy Double Machine Learning to quantify the substantial geographic heterogeneity of climate damages masked by the average (Section 4.2). Finally, we confirm the core non-linear result is highly robust to alternative functional form assumptions (Section 4.3) and the inclusion of precipitation extremes (Section 4.4). (We separately explore the economic mechanisms driving these damages in Section 5).

4.1 The Average Damage Function (Binned Fixed-Effects)

Figure 2 presents the semi-parametric binned damage function. We estimate this by regressing the log GDP level on four non-overlapping monthly temperature bin counts (< 0, 0–10, 10–20, > 20°C), controlling for precipitation, region fixed effects, region-specific linear time trends, and year fixed effects.

These region-specific trends flexibly absorb each region's differential long-run growth trajectory. As detailed in Section 3.1, this specification captures the *level effect* of temperature—isolating transitory disruptions to output volume rather than compounding shocks to the underlying growth rate. The dependent variable is scaled by 100, allowing each coefficient to approximate the percentage-point difference in GDP associated with reallocating one additional month into a given temperature range.

Because all four bins enter the regression simultaneously and the number of months per region-year is approximately fixed, the bins are nearly collinear: adding one month to any bin necessarily removes one month from the remaining bins. As a consequence, the *individual* coefficients are identified only up to a common additive constant; what is well-identified—and economically meaningful—are the *differences* between any two bin coefficients (see Appendix C for the full technical exposition and coefficient table). The

shape and relative ordering of the response curve are therefore robust, even though the absolute vertical position depends on the normalization.

The response curve reveals a "comfort zone" near 10°C. The 0–10°C and 10–20°C bins are statistically indistinguishable (−0.06 pp difference, $t = -0.95$), forming a plateau of optimal temperatures. Reallocations outside this comfort zone yield statistically significant and economically meaningful penalties:

- Reallocating one month from the optimal plateau (using 0–10°C as the reference peak) into the hot bin (> 20°C) is associated with a **0.22 percentage-point loss** in GDP ($p < 0.01$).
- Conversely, moving one month from below 0°C into the optimal range is associated with a **0.52 percentage-point gain**, consistent with the relaxation of cold-induced constraints on agriculture, construction, and energy demand.

The damage pattern is highly asymmetric: the cold penalty (< 0°C vs. 0–10°C: −0.52 pp) exceeds the heat penalty (> 20°C vs. 0–10°C: −0.22 pp), consistent with the disproportionate cold-season constraints on European economies.

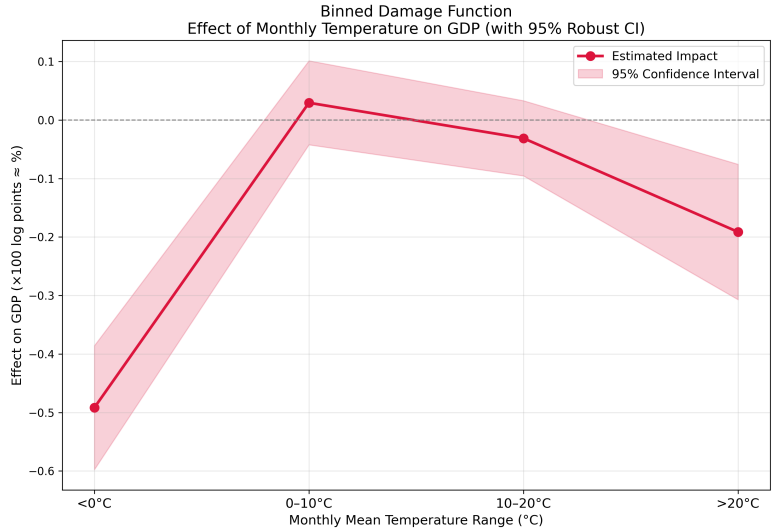


Figure 2: Semi-parametric binned damage function (level specification). Each point shows the estimated effect of one additional month in the given monthly-mean temperature range on $100 \times \ln(\text{GDP})$, controlling for precipitation, region fixed effects, region-specific linear time trends, and year fixed effects. Error bands denote 95% confidence intervals from Conley spatial-HAC standard errors (Bartlett kernel, 500 km cutoff), which account for both within-region serial correlation and cross-regional spatial dependence from contiguous weather systems. The concave shape forms a broad plateau across the 0–20°C ranges, with statistically significant damages for both cold (< 0°C) and hot (> 20°C) months.

4.2 Geographic Heterogeneity (Double Machine Learning)

The binned estimator establishes a highly non-linear temperature–output relationship across Europe. However, because European regions span an interior temperature range (from 0°C in the Nordic countries up to 20°C in Southern Europe) that closely brackets the empirical optimum, the continent-wide average relationship may understate the geographic severity of climate impacts.

We map this geographic inequality using the DML estimator (Section 3.2), which combines causal Neyman-orthogonal residualization with flexible heterogeneity detection. The Conditional Average Treatment Effects (CATE) presented here reflect the unconfounded marginal effect of a +1°C temperature shock for every NUTS-3 region.

4.2.1 Temperature Anomaly (+1°C Shock)

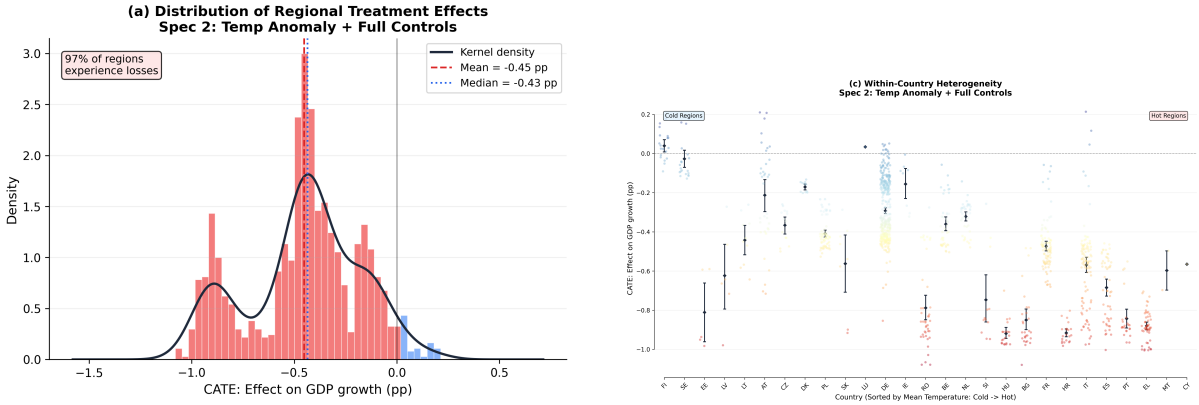
Using within-region temperature anomaly ($T_{it} - \bar{T}_i$) as the treatment removes cross-regional level differences, so the first stage can no longer absorb the treatment through region fixed effects. The aggregate ATE is near zero (-0.0005, $p = 0.576$) and statistically insignificant. For this specification, however, the ATE is not the relevant policy estimand. Because European regions span both sides of the temperature optimum, the continent-wide average mechanically cancels Northern gains against Southern losses. The economically informative objects are the *distribution* of region-specific CATEs and the *group-average treatment effects* (GATEs) across climate zones.

The distributional evidence is systematic. Over 80% of regions display negative CATE point estimates, with a mean of -0.45 percentage points. While individual region-level confidence intervals are wide—a standard feature of Causal Forest estimation at fine geographic resolution—three lines of evidence confirm that this heterogeneity reflects genuine signal rather than noise:

1. **CATE variance test.** The estimated variance of the CATE distribution ($\hat{\sigma}_\tau^2 = 0.18$) substantially exceeds the mean squared estimation error of individual CATEs, indicating that the observed dispersion is not attributable to sampling variability alone (Appendix Table G1).
2. **Group-average contrasts.** Partitioning regions into terciles by baseline temperature, the warm-tercile GATE (-0.74 pp) differs from the cold-tercile GATE (-0.18 pp) by 0.56 pp ($p < 0.01$ via cluster-robust t-test; Appendix Table G1), confirming that the North–South gradient is statistically significant at the group level.
3. **Rank correlation.** The Spearman rank correlation between baseline temperature and estimated CATE is $\rho = -0.75$ ($p < 0.001$), indicating a monotonic, highly systematic geographic gradient (Figure 7).

Nordic countries (Finland, Sweden) cluster near zero; Mediterranean countries (Greece,

Portugal, Spain, Cyprus) face losses of −0.5 to −1.5 pp (Figure 3). Any policy framework that evaluates climate damages at the EU-aggregate level will systematically understate the harm borne by the most vulnerable member states.



(a) Distribution of Region CATEs

(b) Within-Country Heterogeneity

Figure 3: DML Temperature Anomaly (+1°C shock; Appendix G, Spec 2). Panel (a): Over 80% of regions show negative CATEs, with mean = −0.45 pp, despite a non-significant aggregate ATE—reflecting cancellation of Northern gains and Southern losses. Panel (b): countries sorted cold (left) to hot (right).

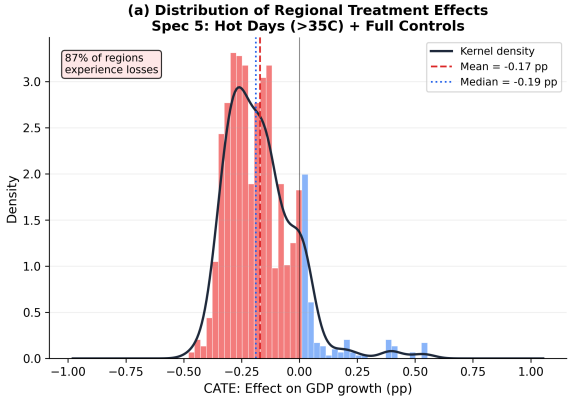
4.2.2 Hot Days > 35°C (+10 Extra Days)

While the temperature-anomaly specification reveals the geography of vulnerability, it does not achieve formal statistical significance at the aggregate level. The hot-days specification addresses this by defining the treatment as the count of days per year where the maximum temperature exceeds 35°C—isolating the impact of *extreme heat events* rather than average temperature shifts. This treatment definition is mechanistically precise: heat waves are the primary driver of labor productivity collapse, crop failure, and excess mortality [Carleton and Hsiang, 2016, Dell et al., 2014, Usman et al., 2025].

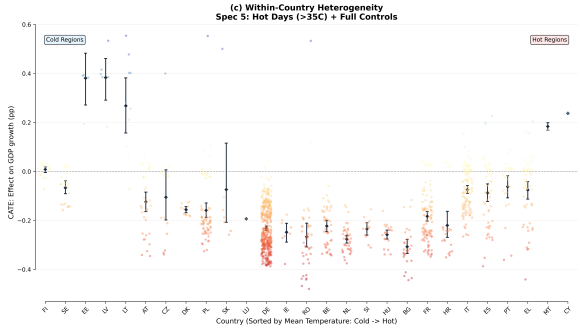
The results are statistically significant: the ATE is −0.00012 ($p = 0.010$), making this the only DML specification to achieve significance at the 1% level. To translate this into economically interpretable magnitudes: because the dependent variable is $100 \times \Delta \ln Y$, the ATE of −0.00012 per additional hot day implies that a shock of 10 extra days above 35°C reduces GDP growth by approximately 0.12 percentage points.[2] The distribution of region-specific CATEs shows 87% of regions with negative effects (mean = −0.17 pp per 10 extra hot days; Figure 4). The strip chart reveals a clear structural divergence: Baltic/Nordic states show *positive* CATEs because they rarely experience days above

[2]The raw ATE is reported in units of the dependent variable ($100 \times \Delta \ln \text{GDP}$) per unit of treatment (one additional day > 35°C). Multiplying by 10 days gives the per-decade-equivalent shock. Region-level CATEs reported throughout Section 4.2 are pre-multiplied by the treatment dose stated in each subsection header (e.g., +10 days, +10 CDD, +1°C anomaly) and are thus directly in percentage points of GDP growth.

35°C and the counterfactual shock has little bite, while Southern European countries show uniformly negative effects.



(a) Distribution of Region CATEs



(b) Within-Country Heterogeneity

Figure 4: DML Hot Days > 35°C (+10 extra days/year; Appendix G, Spec 5). Panel (a): 87% of regions show negative effects; ATE = −0.00012 (p = 0.010). Panel (b): Baltic/Nordic states show positive CATEs; Southern Europe shows uniformly negative effects.

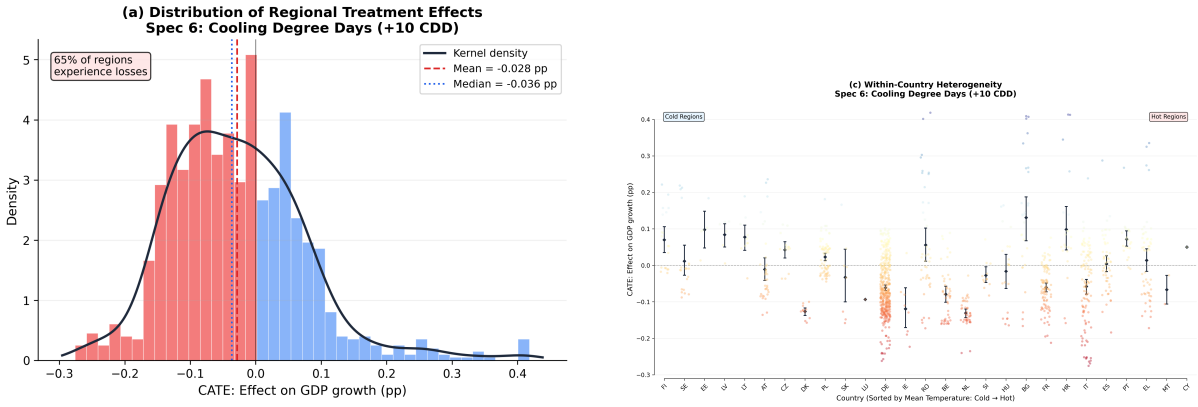
4.2.3 Cooling Degree Days (+10 CDD)

Cooling degree days (CDD) provide a continuous, degree-weighted measure of cumulative thermal burden—capturing not just whether a day exceeds a threshold, but by *how much*. This makes CDD a natural bridge between the discrete hot-days count (Spec 5) and the smooth temperature anomaly (Spec 2), and it maps directly to energy demand for air conditioning and workplace cooling.

The DML results confirm CDD's causal relevance: the ATE is −0.000049 per CDD unit ($p = 0.022$), making this one of only three specifications to achieve statistical significance at the 5% level. We evaluate the CATE at a +10 CDD shock—roughly 10% of the sample mean ($\overline{\text{CDD}} = 106$) and 0.87 standard deviations of the residualized treatment variation, ensuring the counterfactual lies within the dense support of the identifying variation. The CATE distribution (Figure 5) is predominantly negative: the mean CATE is −0.03 pp, with 65% of regions experiencing economic losses from the increased thermal burden.

The strip chart reveals a coherent geography of vulnerability. The hardest-hit countries are moderate-climate economies—Denmark, Ireland, the Netherlands, Slovenia, Belgium—with mean CATEs in the range −0.08 to −0.13 pp. Many Southern European countries (Spain, Greece, Portugal) and the Baltics/Nordics (Finland, Estonia, Latvia, Lithuania) show mildly positive marginal effects. This pattern is consistent with an adaptation interpretation: regions with historically high CDD accumulations may have invested in cooling infrastructure and adjusted work schedules over time, muting the

marginal damage of an additional 10 CDD—even though these same regions suffer the largest *overall* damages from warming (as shown by the temperature-anomaly specification). In contrast, moderate-climate regions that are newly experiencing sustained heat stress—but lack commensurate adaptive capacity—bear the largest marginal damages from incremental extreme-heat exposure. This complements the hot-days finding (Spec 5) by demonstrating that vulnerability depends critically on the gap between current *cumulative thermal load* and historical adaptive capacity.



(a) Distribution of Region CATEs (b) Within-Country Heterogeneity

Figure 5: DML Cooling Degree Days (+10 CDD; Appendix G, Spec 6). Panel (a): 65% of regions show negative CATEs; ATE = −0.000049 per CDD ($p = 0.022$). Panel (b): countries sorted cold (left) to hot (right). Moderate-climate countries (Denmark, Ireland, Netherlands, Belgium) show the largest losses, consistent with rising CDD exposure without commensurate adaptation infrastructure.

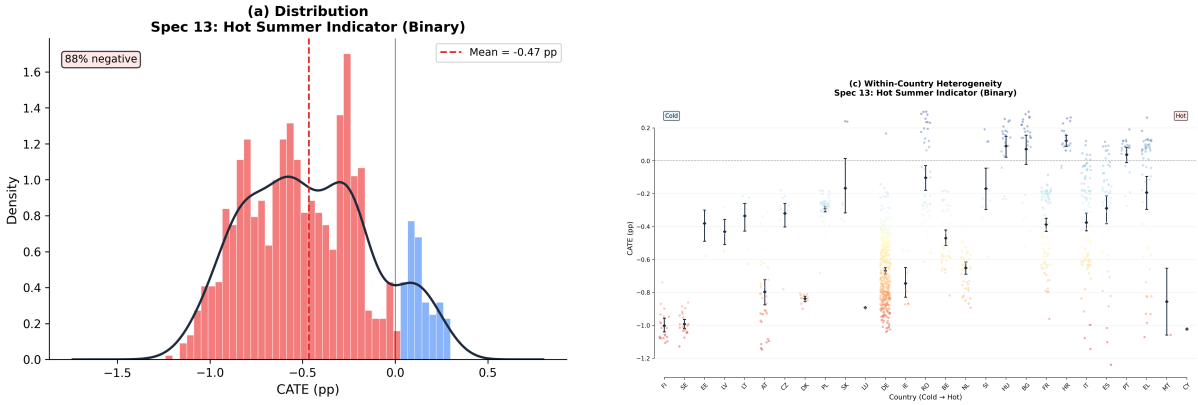
4.2.4 Hot Summer Indicator (Binary)

As a further check, we define the treatment as a binary indicator for whether a region's mean summer temperature exceeds the 75th percentile of its historical distribution. The result is again significant: ATE = −0.0033 ($p = 0.021$), with 88% of regions showing negative CATEs and a mean of −0.47 pp. The strip chart (Figure 6) closely mirrors the hot-days patterns, confirming that the signal is robust to the precise definition of extreme heat.

4.2.5 The Geography of Vulnerability

Figure 7 plots the DML-estimated CATEs against baseline temperature for both the temperature-anomaly and hot-days specifications. The two panels tell complementary but distinct stories about the geography of climate vulnerability.

The temperature-anomaly panel (left) reveals a steep, monotonic gradient: hotter baseline regions suffer systematically larger damages. The slope is −0.069 pp/°C, and the rank correlation is striking (Pearson $r = -0.70$, Spearman $\rho = -0.75$, both $p <$



(a) Distribution of Region CATEs (b) Within-Country Heterogeneity

Figure 6: DML Hot Summer Indicator (binary; Appendix G, Spec 13). Panel (a): 88% of regions show negative effects, with mean CATE = −0.47 pp. Panel (b): the country-level gradient confirms the North–South divergence is robust to treatment definition.

0.001). Regions with baseline temperatures below ∼5°C—predominantly in Scandinavia and the Baltics—cluster near zero or show mildly positive effects, consistent with warming relaxing cold-season constraints. Above ∼10°C, damages steepen rapidly, reaching −0.5 to −1.0 pp in the warmest Mediterranean regions. This gradient is what makes the continent-wide ATE so misleading: Northern gains and Southern losses nearly cancel in aggregate.

The hot-days panel (right) displays a different structure. The regression slope is essentially flat (+0.004 pp/°C; Spearman $\rho = -0.016$, $p = 0.60$), yet the vertical dispersion is enormous—particularly among moderate-temperature regions (7–12°C), where CATEs range from +0.4 to −0.4 pp. This pattern is consistent with a joint exposure-adaptation mechanism: cold regions rarely experience days above 35°C, so the counterfactual shock has little bite; traditionally warm regions (the Mediterranean), despite suffering the largest overall damages from warming, may have built some adaptive capacity against incremental extreme-heat events (workplace cooling, adjusted schedules, heat-tolerant agricultural practices); moderate regions, however, face rising exposure to extreme-heat events under climate change without the commensurate adaptation infrastructure, leaving them most vulnerable.

The subgroup analysis (Figure 8) sharpens these patterns along two dimensions. By climate zone, the temperature-anomaly specification shows a steep escalation: cold regions lose −0.18 pp, moderate regions −0.40 pp, and warm regions −0.74 pp—a fourfold increase from coldest to warmest. The hot-days specification, by contrast, shows a non-monotonic pattern: moderate-climate regions bear the largest damages (−0.28 pp), exceeding both cold (−0.11 pp) and warm (−0.13 pp) zones. This reinforces the exposure-adaptation interpretation: cold regions lack exposure, warm regions—though most damaged overall—may have built some resilience against *marginal* extreme-

heat events through long historical experience, and moderate regions sit in a vulnerable middle ground—increasingly exposed but insufficiently adapted.

By income group, the temperature-anomaly specification reveals a stark inequality: low-income regions lose −0.61 pp—roughly double the damage borne by mid-high-income regions (−0.29 pp). Under the hot-days specification, the income gradient is flatter (−0.15 to −0.22 pp), suggesting that the protective role of income is more pronounced against broad warming trends than against acute heat extremes. Taken together, these patterns indicate that the largest damages concentrate in warm, lower-income regions—precisely the areas with the fewest resources for adaptation—while vulnerability to acute heat events is more diffuse and less predictable from baseline characteristics alone.

This exposure-adaptation gap resolves a persistent conceptual tension in climate economics: if Southern Europe has been hot for centuries, why do they suffer the largest aggregate damages from warming, rather than showing the strongest adaptation? The answer lies in the distinction between a marginal economic shock and an absolute physiological ceiling. Because the underlying biological damage functions of crop yields and human heat stress are exponentially non-linear, a +1°C anomaly in the Mediterranean translates into absolute temperatures that breach physiological limits. The same +1°C anomaly in Northern Europe yields absolute temperatures that remain within comfortable, productive bounds. Thus, while Southern Europe is indeed adapted to their high historically warm baseline, they are operating so close to the absolute physical limits of thermal tolerance that a marginal +1°C anomaly pushes them "over the cliff," overwhelming their defensive investments. We formally validate these gradients using alternative functional forms in Section 4.3.

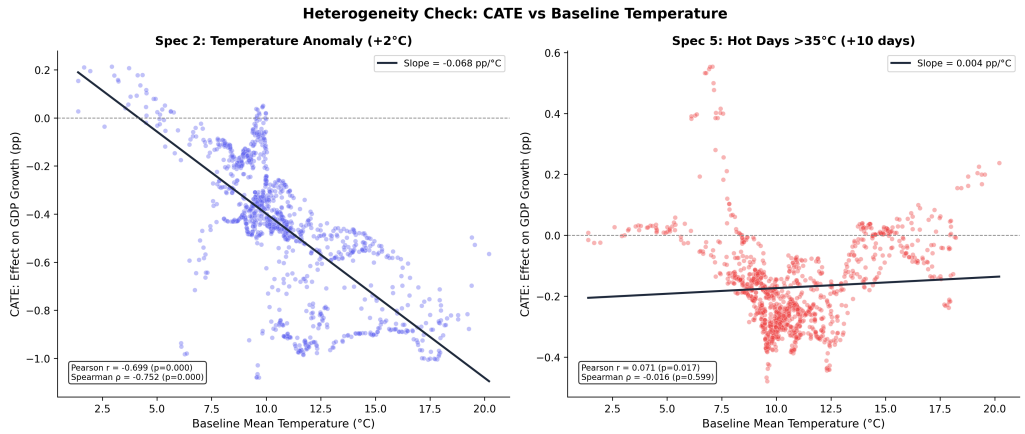


Figure 7: DML CATEs vs. baseline temperature for the temperature-anomaly and hot-days specifications. The downward slope confirms that hotter baseline regions suffer larger negative impacts (a gradient confirmed via alternative methods in Section 4.3). Pearson and Spearman correlations are reported in each panel.

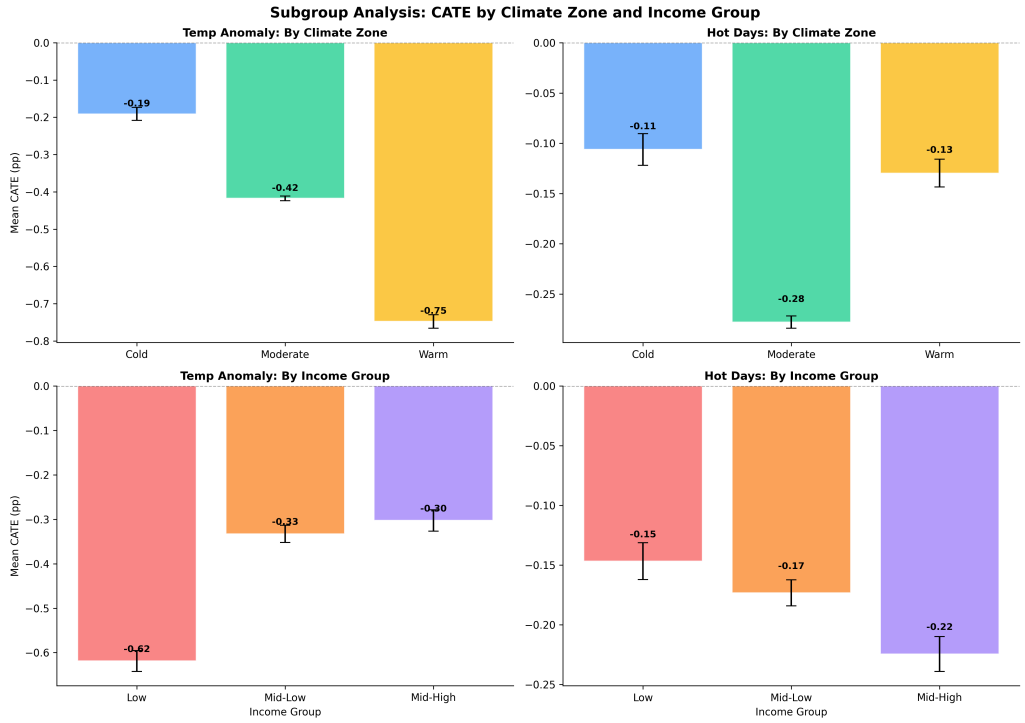


Figure 8: Subgroup analysis of DML CATEs by climate zone and income group, for the temperature-anomaly and hot-days specifications. The largest damages concentrate in warm and hot climate zones, while higher-income regions show somewhat more moderate effects, consistent with greater adaptive capacity.

4.3 Robustness to Functional Form

Our core result—a highly non-linear damage function that masks substantial interregional heterogeneity—was estimated using two flexible methods. To ensure these findings are not artifacts of subjective researcher choices (e.g., bin cutoffs, random forest hyperparameters), we validate them here using entirely independent functional forms.

1. The Parametric Verification. Our parametric panel fixed-effects models (reported in full in Appendix B, Table B1) confirm the concave relationship. A classic quadratic specification yields a significant negative coefficient on T^2 ($\hat{\beta}_{T^2} = -0.007$, $p < 0.01$). To verify the heterogeneity found by the DML, we estimate a simple linear interaction model: $\log \text{GDP}_{it} = \beta_1 \cdot \text{TempAnomaly}_{it} + \beta_2 \cdot (\text{TempAnomaly}_{it} \times \overline{T}_i) + \gamma' \mathbf{X}_{it} + \varepsilon_{it}$. The interaction term $\hat{\beta}_2$ is strongly negative (≈ -0.55), exactly corroborating the DML finding that identically sized heat shocks cause larger damages in hotter regions. Figure 9 plots this parametric marginal effect line: cold regions correctly show positive marginal effects, while hot regions show steep damages.

2. The Continuous GAM Verification. To validate the discrete binned results continuously, we estimate the penalized-spline GAM on double-demeaned region–year observations (DV = GDP growth rate). This non-parametric approach smooths through the data without assuming a polynomial curvature. Figure 10 confirms the concave

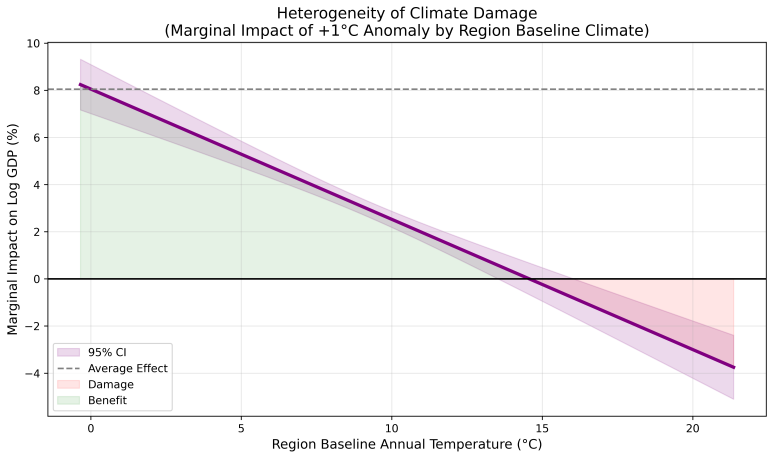


Figure 9: Robustness Check: Parametric Interaction Model. The marginal effect is $\hat{\beta}_1 + \hat{\beta}_2 \times \overline{T}_i$, with 95% confidence interval (purple band) from cluster-robust standard errors. This highly restrictive parametric form precisely recovers the same North-South inequality identified by the flexible DML estimator.

shape identified by the parametric and binned estimators, placing the optimum near 10°C (slightly below the parametric crossover of 11.0–11.4°C, consistent with the flexible spline capturing a broader plateau) and confirming that the curvature at the peak is significantly negative ($p < 0.001$).

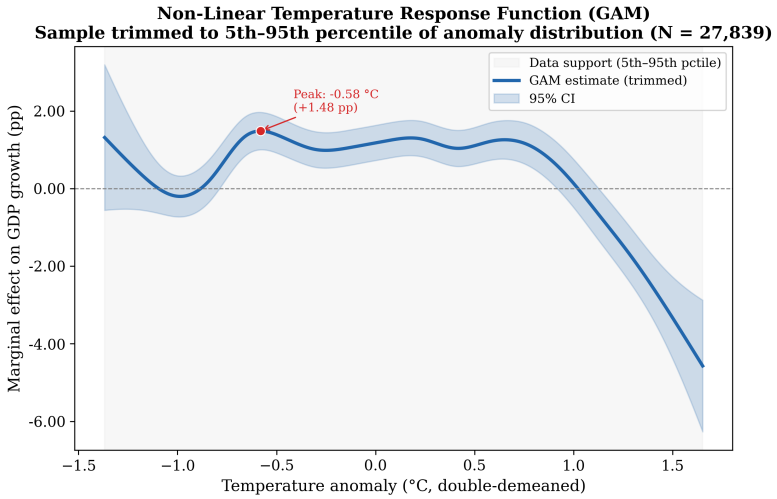


Figure 10: Robustness Check: Continuous GAM Response Function. The smoothly estimated B-spline confirms the exact peak ($\sim$ 10°C) and non-linear shape identified by the bins and the parametric models. Dependent variable is double-demeaned GDP growth.

3. Split-Sample Stability. Finally, we investigate whether the temperature–growth relationship has evolved over time by estimating our core parametric model on two split samples: an early period (1995–2008) and a late period (2009–2022). We find that the non-linear, concave relationship persists firmly across both periods ($p < 0.05$ for the quadratic temperature term in both sub-samples). However, the magnitude of the

temperature penalty appears somewhat attenuated in the later period ($\hat{\beta}_{T^2} = -0.027$ for 2009–2022 versus $\hat{\beta}_{T^2} = -0.047$ for 1995–2008). This attenuation is suggestive of gradual regional adaptation to heat stress, though the persistence of significant quadratic damages even in the recent decade underscores that adaptation remains incomplete.

4.4 Joint Temperature–Precipitation Effects

Temperature does not operate in isolation. Our parametric results consistently show that precipitation anomalies—both floods and droughts—exert statistically significant negative effects on regional growth [Kotz et al., 2022]. The damage function is concave: extreme hydrological events cause disproportionate losses compared with mild fluctuations. A comprehensive analysis of precipitation transmission mechanisms, including a formal mediation decomposition showing that floods operate primarily through capital and productivity destruction while droughts trigger labor-market contraction, is presented in Appendix F.

5 Economic Mechanisms

Having established that hot southern regions bear the brunt of climate damages, we turn to the question of *why*. We decompose the net growth effect for the hottest 10% of regions—those situated firmly in the damage zone—into suggestive contributions from four economic channels: labor productivity, investment, employment, and capital stock (Figure 11). The decomposition follows a product-of-coefficients mediation framework with delta-method (Sobel) standard errors for every indirect effect and cluster-robust standard errors for the total and direct effects, ensuring that each channel's contribution can be assessed for statistical significance. To address potential violations of sequential ignorability from region-specific trends (e.g., differential catch-up growth), the preferred specification absorbs region-specific linear time trends via Frisch–Waugh–Lovell projection, in addition to the standard region and year fixed effects. We emphasize that this decomposition is informative about the relative importance of economic channels (Section 3.4), rather than a definitive structural causal partition.

The total effect of a unit temperature anomaly on GDP growth in the hottest decile is -0.582 percentage points (SE $= 0.191$, $p < 0.01$). Under the identifying assumptions of the mediation framework (Section 3.4), the decomposition points to a single dominant channel. Labor productivity alone contributes -0.558 pp to the total effect (Sobel SE $= 0.124$, $p < 0.001$), accounting for 95.9% of the damage. In terms of a standard Cobb-Douglas production function $Y = A \cdot K^{\alpha} L^{1-\alpha}$, this result implies that temperature operates overwhelmingly through the TFP channel (A) rather than through factor accumulation (K, L)—heat degrades the efficiency with which inputs are combined, not

the quantity of inputs deployed. If one could fully insulate labor productivity from temperature, the estimated net effect on GDP growth would be near zero. The tight 95% confidence interval [−0.801, −0.315] suggests that this channel is the primary driver of the observed damage. This finding is consistent with the physiological channel of climate damage: extreme heat degrades physical and cognitive capacity, reducing output per hour worked, particularly in exposed sectors such as agriculture and construction [Zhang et al., 2024, Yu and Shi, 2025, Mérel and Gammans, 2021]. The workforce is not shrinking; it is becoming less efficient—a distinction with important policy implications, since employment-focused interventions may miss the binding constraint.

Investment contraction contributes approximately −0.116 pp (SE = 0.078, 20.0% of the total), suggesting a meaningful economic channel. However, the Sobel test does not reach conventional significance ($p = 0.136$), reflecting substantial uncertainty in either the treatment→mediator or mediator→outcome pathway. Capital stock depreciation adds a further −0.231 pp (SE = 0.134, $p = 0.084$), which approaches marginal significance when trends are absorbed.

Employment, measured here as the total headcount of persons employed (the extensive margin), perhaps counter-intuitively, makes a *positive* and highly significant contribution of +0.235 pp (SE = 0.059, $p < 0.001$), offsetting 40.4% of the total damage under our preferred specification. The narrow 95% CI [+0.119, +0.351] provides strong statistical support that this buffering channel operates meaningfully in the aggregate. However, rather than reflecting localized, within-region factor substitution by highly constrained individual firms, this employment expansion operates as a macro-institutional phenomenon. Because regional data typically capture the extensive employment margin rather than total hours worked (the intensive margin), we cannot definitively rule out that existing employees also work longer hours to compensate. Yet, as explored below, this robust extensive-margin expansion is fundamentally tied to national-level labor market dynamics, providing a partial aggregate buffer that is nevertheless insufficient to offset the total productivity collapse. This result is robust to the inclusion of region-specific linear time trends (Table D2), but dissipates entirely under an absorbing specification with country×year fixed effects (Appendix D.4).

The residual net direct effect is +0.089 pp (SE = 0.076), small and statistically insignificant at the 5% level, indicating that the four mediators collectively account for the vast majority of the total effect. The near-zero residual suggests that we have captured the primary economic mechanisms driving temperature damages in these regions.

Sectoral regressions (Appendix E) reveal where these losses concentrate. In the hottest regions, agriculture and services suffer significant damages at +1°C, while manufacturing remains unaffected. Construction shows a positive response to warming—consistent with reconstruction demand rather than genuine productivity gains. While agriculture suffers severe percentage losses, it represents too small a share of baseline regional GDP (< 5%)

to mathematically drive the aggregate −0.6 pp macroeconomic penalty alone. The sheer magnitude of the total effect is instead driven by the services sector—which constitutes over 70% of the Mediterranean economy—where extreme heat suppresses tourism demand, degrades logistics, and impairs cognitive productivity. These sectoral findings confirm that the aggregate productivity channel is a composite of outdoor exposure, labor intensity, and sector-specific adaptation capacity.

Together, the mediation results indicate a fundamental productivity crisis rather than a demand-side contraction. Effective adaptation in hot regions requires interventions that specifically decouple labor productivity from thermal stress: workplace cooling, schedule adjustments, and sector-specific heat protocols. Crucially, the aggregate magnitude of these damages highlights the physical and economic limits of conventional adaptation, helping explain the paradox of why historically warm regions remain so vulnerable. While capital investments such as air conditioning can protect indoor manufacturing and services, the most exposed sectors—agriculture and construction—are inherently outdoor activities where the scope for mechanized cooling is fundamentally bounded [Zhang et al., 2024]. This explains why Mediterranean firms cannot simply "buy more air conditioning" to fully insulate their workers from warming, leaving them exposed to the exponential damages of severe heat.

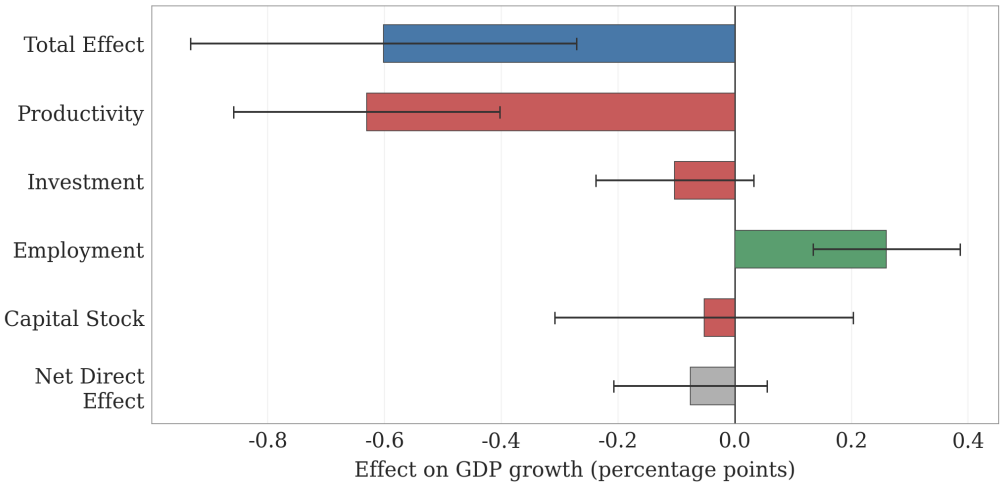


Figure 11: Mediation decomposition of the total temperature effect for the hottest decile of European regions (preferred specification with region-specific linear time trends). Bars show point estimates; whiskers denote 95% confidence intervals. Red bars indicate damage channels; green indicates a buffering channel; grey is the residual direct effect. The total effect is −0.582 pp (SE = 0.191, $p < 0.01$). Productivity dominates, contributing −0.558 pp (SE = 0.124, $p < 0.001$; 96% of total). Employment expansion provides a significant but insufficient offset of +0.235 pp (SE = 0.059, $p < 0.001$; −40%). Investment (−0.116 pp, SE = 0.078, $p = 0.136$) and capital stock (−0.231 pp, SE = 0.134, $p = 0.084$) are not individually significant at the 5% level. Indirect-effect SEs computed via the Sobel delta method.

6 Conclusion

Five independent estimation strategies applied to 1,379 European regions converge on a consistent conclusion: economic growth follows a concave, non-linear relationship with temperature, peaking near 11°C. The distributional damages are tangible, non-linear, and geographically concentrated—even as the continent-wide average effect is near zero.

6.0.1 Welfare Translation

To translate our preferred estimates into economically interpretable magnitudes, Table 3 reports the implied annual GDP growth losses under three warming scenarios spanning the range of current climate projections.

Table 3: Welfare Translation: Implied Annual GDP Growth Effects by Region Climate Zone

Region Group	**+1°C (SSP1-2.6)**	**+1.5°C (SSP2-4.5)**	**+2°C (SSP3-7.0/RCP 8.5)**	**Scenario Reference**
Coldest quintile (Nordic/Baltic)	+0.05 pp	+0.08 pp	+0.10 pp	Mild gain
Cold-moderate (N. Central)	−0.18 pp	−0.27 pp	−0.36 pp	Small loss
Moderate (Central)	−0.40 pp	−0.60 pp	−0.80 pp	Moderate loss
Warm-moderate (S. Central)	−0.58 pp	−0.87 pp	−1.16 pp	Substantial loss
Warmest quintile (Mediterranean)	−0.74 pp	−1.11 pp	−1.48 pp	Severe loss
EU-wide ATE (Causal Forest)	≈ 0 pp	≈ 0 pp	≈ 0 pp	Near-zero (redistribution)

Notes: Growth effects computed by scaling the DML group-average treatment effects (GATEs) from the temperature-anomaly specification (Spec 2) by the indicated warming increment. The coldest quintile (baseline temperature < 7°C; Scandinavia, Baltics) shows mildly positive GATEs, consistent with warming relaxing cold-season constraints. The Causal Forest ATE (-0.0005 pp/°C, $p = 0.576$) is approximately zero because gains in the coldest quintile partially offset losses in the warmer four-fifths; the residual is economically negligible (< 0.001 pp) and statistically indistinguishable from zero. SSP scenario mapping follows IPCC AR6 WG1 median projections for European mean temperature by 2050 relative to the 1995–2014 baseline. These are contemporaneous level effects; cumulative welfare losses depend on persistence assumptions not estimated here.

Three clear policy implications emerge. First, for Southern Europe, climate damage presents a pressing productivity challenge rather than a distant risk. Declining labor productivity accounts for the majority of economic losses from heat, outweighing compensatory employment responses. Under a moderate warming trajectory (SSP2-4.5, +1.5°C by 2050), Mediterranean regions face annual growth losses exceeding 1 percentage point; under SSP3-7.0 (+2°C), losses approach 1.5 pp. Even under the optimistic SSP1-2.6 pathway (+1°C), the warmest quintile loses 0.74 pp annually. Adaptation policies should therefore prioritize workers directly—through workplace cooling, shifted schedules, and heat-safety protocols—rather than relying solely on general employment support. For agriculture, accelerating capital substitution through mechanization offers a structural defense, though this remains challenging for the fragmented, horticulture-based farm structures typical of Southern Europe. The EU Mission on Adaptation to

Climate Change, the 2024 revision of the European Climate Adaptation Strategy, and the climate-mainstreaming provisions of the European Green Deal provide institutional frameworks for channeling such investments toward the most exposed regions. The Common Agricultural Policy (CAP) should similarly prioritize climate-resilience measures in Mediterranean member states, where our sectoral analysis (Appendix E) reveals that agriculture faces compounded triple exposure to heat stress, drought, and flood risk—each operating through distinct transmission channels (Appendix F).

Second, relying on European averages is misleading. While the EU-wide ATE is approximately zero across all scenarios (Table 3), this aggregate conceals a spatial divergence of roughly 1 percentage point of annual growth between the South and the North. EU Cohesion Policy—particularly the European Regional Development Fund (ERDF) and the Just Transition Fund—must be recalibrated to recognize climate vulnerability as a primary geographic economic constraint, complementing the objectives set out in the European Climate Law (Regulation 2021/1119). Climate-weighted allocation formulas that incorporate regional heat exposure alongside traditional GDP-per-capita criteria would better target resources toward the regions bearing the greatest damages.

Third, the relative insulation of Northern Europe from warming effects may be transient. With the economic optimum effectively at 11°C, many northern regions are already operating near the limit of their benefit zone. As temperatures rise further, the area of economic impact is likely to expand northward.

6.1 Limitations and Generalizability

While our causal estimation framework provides robust identification of short-run climate impacts across Europe, several limitations constrain the interpretation of these results. First, because our panel fixed-effects approach relies on high-frequency, year-to-year weather shocks to achieve causal identification, the resulting estimates capture an immediate level effect rather than a compounding growth effect. As such, these short-run damages likely represent a lower bound on long-run climate costs if growth effects dominate, or an upper bound if adaptation attenuates level effects over time [Kahn et al., 2021]. Recent work using global temperature variation [Bilal and Känzig, 2026] estimates damages 6–10 times larger than local-panel studies, suggesting that general-equilibrium amplification—through trade disruption, correlated global shocks, and aggregate demand effects—may substantially magnify the partial-equilibrium, region-specific effects estimated here. Over decadal horizons, regions can adapt through directed capital investments (e.g., HVAC deployment, resilient infrastructure) and structural transformation that annual shock models cannot seamlessly capture [Heutel et al., 2021, Kahn et al., 2021].

Second, the structural mediation framework relies on the assumption of sequential

ignorability. While we condition on a rigorous set of spatial and temporal fixed effects alongside climate controls, we cannot definitively rule out unobserved, time-varying confounders that might jointly determine mediator and outcome growth. To address this concern, our preferred mediation specification absorbs region-specific linear time trends via Frisch–Waugh–Lovell projection (Table D1). When we compare this to the baseline without trends (Table D2), the channel decomposition is qualitatively unchanged: the labor-productivity share increases from 83% to 96%, and the employment buffer remains significant, confirming that the mediation results are not driven by linear confounders. As an additional sensitivity test, replacing year fixed effects with country×year fixed effects (Appendix D.4) strengthens the total effect from -0.582 to -0.897 pp, but the severely reduced within-country identifying variation (8 countries in the hottest decile) renders all individual channel estimates imprecise. Consequently, the mediation results represent an informative and robust decomposition of the reduced-form effect—the productivity dominance and employment buffering are stable across the preferred specification and the trends robustness check, though the channel-level decomposition cannot be precisely identified in the most restrictive fixed-effects structure. Furthermore, while structural macro-models (such as CGE frameworks frequently used in institutional climate assessments) often emphasize capital depreciation and transition risks while downplaying labor productivity, our empirical ex-post estimates capture all compounding behavioral and economic frictions on the ground, including cognitive impairments and morbidity that strictly physiological models tend to miss.

Third, our findings are specific to the European context. Europe benefits from robust institutions, high baseline wealth, and substantial integration, which may facilitate factor substitution (such as the observed employment buffering) more effectively than would be possible in developing economies. Conversely, Europe's relatively moderate baseline climate means extreme heat events may trigger larger percentage losses here than in regions already historically adapted to intense heat. Extrapolating the precise magnitude of these damages to other continents requires caution.

Future research could productively extend this analysis in several directions. First, distinguishing between physical heat stress and cognitive impairment requires finer-grained microdata at the firm or worker level. Second, given the localized nature of these damages, investigating how climate shocks propagate across regions through trade and supply chain linkages—as recently demonstrated at the country level [Schenker and Osberghaus, 2025]—would provide a more complete picture of aggregate economic vulnerability. Third, implementing a permutation-based placebo test—randomly reassigning temperature realizations across regions within each year—would provide a non-parametric benchmark for the borderline placebo result ($p = 0.061$) discussed in Section 3.3, strengthening (or qualifying) the causal interpretation.

However, while these limitations nuance the interpretation, the robust agreement

across multiple identification strategies remains clear: unmitigated temperature is a binding constraint on regional economic performance. The short-run economic costs are substantial, and they fall hardest on the regions least equipped to bear them.

Data Availability Statement

All data used in this study are drawn from publicly available sources: regional economic accounts from Eurostat, climate variables from the Copernicus ERA5 reanalysis, and the harmonized panel of Usman et al. [2025]. A complete replication package will be made available upon acceptance. The replication materials are available to editors and referees upon request during the review process.

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Online Appendix

A Comparison of Econometric Specifications

Table A1: Comparison of Econometric Specifications

	(1) Parametric	(2) Binned FE	(3) GAM	(4) DML
Model	Quadratic Panel FE	Binned Panel FE	Penalized Spline	DML Causal Forest
Dep. Variable	$100 \times \Delta \ln Y$ (growth, %)	$100 \times \ln Y$ (log level)	$\widetilde{\Delta \ln Y}$ (demeaned growth)	$100 \times \Delta \ln Y$ (growth, %)
Functional Form	$\beta_1 T + \beta_2 T^2$	$\sum \beta_j \text{Bin}_j$	$s(T; \lambda)$	Two-stage residuals
Non-Linearity	Imposed (Convex)	Flexible (Stepwise)	Smooth (Spline)	Data-Adaptive
Heterogeneity	Interaction Only	–	Stratified	Fully Heterogeneous
Causal ID	Panel FE	Panel FE	Pre-demeaning	Neyman-orthogonal
Controls ($\mathbf{Z}_{it}$)				
Lagged GDP	–	–	–	Cross-fitted
Precipitation	Yes	Yes	Demeaned	Cross-fitted
Region-Specific Trends ($\delta_i \cdot t$)	–	Yes	–	–
Fixed Effects				
Region (μ_i)	Yes	Yes	Pre-demeaned	Cross-fitted
Year (θ_t)	Yes	Yes	Pre-demeaned	Cross-fitted
Inference	Cluster-Robust SE	Cluster-Robust SE	Bayesian CI	Neyman-orthogonal

Notes: $100 \times \Delta \ln Y$ denotes GDP growth (the log-difference of real GDP, scaled by 100 so coefficients read as percentage points); $100 \times \ln Y$ denotes the log GDP level (scaled by 100 for readability); $\widetilde{\Delta \ln Y}$ denotes GDP growth double-demeaned by region and year. The parametric models (Table B1) and DML use GDP *growth* (scaled by 100) as the dependent variable; the binned model uses the log GDP *level*; the GAM uses double-demeaned growth. All specifications exploit within-region variation after absorbing region and year fixed effects. The GAM pre-demeans both the outcome and treatment by region and year before estimation. The DML uses cross-fitted gradient-boosted residualization to ensure Neyman-orthogonality, then estimates heterogeneous effects via Causal Forest. "Cross-fitted" indicates that controls are absorbed via 5-fold GroupKFold cross-fitting in the first stage.

B Parametric Estimation Details

B.1 Specification Progression

Table B1 reports a sequence of two-way fixed-effects models that progressively relax restrictions on how temperature enters the growth equation. The progression reveals why simple models fail to capture the European temperature–growth relationship:

- **Column 1 (Levels only):** The canonical quadratic ($T + T^2$) yields a negative but insignificant T^2 coefficient (-0.004, $p > 0.05$), providing only weak evidence for the concave relationship.
- **Columns 2–3 (Pooled anomaly):** Adding a single "temperature anomaly" term produces a marginally significant negative effect in Column 2 (-0.124, $p < 0.05$), but pooling warming and cooling shocks cancels their distinct effects. The anomaly $\times$ baseline interaction in Column 3 is insignificant.

- **Column 4 (Split anomaly):** The signal crystallizes when anomalies are decomposed: cooling anomalies are significantly positive (+0.352, $p < 0.01$), while warming anomalies are positive but insignificant (+0.086).
- **Columns 5–6 (Interactions):** Interacting directional anomalies with baseline climate reveals the key heterogeneity. In the full specification (Column 6), both warming and cooling main effects are large and positive (+1.20, +2.16; both $p < 0.001$), but the interactions with baseline temperature are strongly negative (−0.109, −0.189; both $p < 0.001$).

Across Columns 3–6, precipitation extremes consistently reduce growth: flood anomalies by ≈ 0.027–0.029 pp per mm excess ($p < 0.001$), and drought anomalies by ≈ 0.011–0.013 pp per mm deficit ($p < 0.05$).

B.2 Interpreting the Interaction Structure

In Columns 5–6 of Table B1, both warming (+1.20) and cooling (+2.16) main effects are positive. This does *not* mean that all temperature shocks are beneficial. The main effects give the impact at a hypothetical baseline of $\bar{T}_i = 0$°C—well outside the European data range. The economically relevant quantity is the *total marginal effect*, which combines the main effect with the interaction:

$$\frac{\partial\,\text{Growth}}{\partial\,\text{Warming}} = 1.20 - 0.109\,\bar{T}_i, \qquad \frac{\partial\,\text{Growth}}{\partial\,\text{Cooling}} = 2.16 - 0.189\,\bar{T}_i$$

Table B2 evaluates these at representative baselines. The crossover occurs near 11°C for both shock types—closely matching the damage-function optimum from the GAM and binned estimators.

C Binned Estimator: Methodology and Interpretation

This section provides the full technical exposition of the binned damage function reported in Figure 2 of the main text.

C.1 Bin Construction

For each region i and month m in year t, we observe the monthly mean temperature $T_{i,m,t}$. Each month is assigned to one of four non-overlapping bins:

$$\textsc{Bin}_k = \{m : T_{i,m,t} \in [\tau_k, \tau_{k+1})\}, \quad k = 1, \ldots, 4$$

Table B1: Parametric Estimation Results: Temperature and Precipitation Effects on GDP Growth

Dependent variable: $100 \times \Delta \ln Y$ (GDP growth in percentage points)						
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Temperature						
Temperature	0.199*	0.185*	0.213**	0.198*		0.328***
	(0.088)	(0.076)	(0.081)	(0.077)		(0.077)
Temperature^2	−0.004	−0.002	−0.004	−0.003		−0.011*
	(0.005)	(0.005)	(0.006)	(0.005)		(0.005)
Temp Anomaly		−0.124*	−0.211			
		(0.061)	(0.155)			
Interaction (Anom × Level)			0.010			
			(0.017)			
Temp Anom Warming				0.086	1.408***	1.203***
				(0.104)	(0.229)	(0.238)
Temp Anom Cooling				0.352**	1.806***	2.156***
				(0.109)	(0.259)	(0.273)
Interaction (Warm × Level)					−0.125***	−0.109***
					(0.022)	(0.024)
Interaction (Cool × Level)					−0.166***	−0.189***
					(0.027)	(0.029)
Precipitation						
Precipitation	0.000					
	(0.001)					
Precipitation^2	−0.000					
	(0.000)					
Precip Anomaly		−0.012***				
		(0.002)				
Precip Anom Flood			−0.029***	−0.028***	−0.027***	−0.027***
			(0.003)	(0.003)	(0.003)	(0.003)
Precip Anom Drought			−0.011*	−0.012*	−0.013*	−0.013*
			(0.005)	(0.005)	(0.005)	(0.005)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31,622	31,622	31,622	31,622	31,622	31,622
Overall R^2	0.242	0.243	0.244	0.244	0.244	0.245
Within R^2	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01

Notes: Cluster-robust standard errors in parentheses (clustered at the region level). All directional anomaly variables—warming, cooling, flood, and drought—are expressed as positive magnitudes (e.g., "Temp Anom Cooling" equals $|\min(T_{it} - \bar{T}_i, 0)|$), so each coefficient gives the effect of a one-unit *increase* in the magnitude of that type of shock. Column 1: baseline quadratic in temperature levels. Columns 2–3: pooled anomaly and unsplit interaction. Column 4: anomalies split by direction (warming/cooling). Column 5: directional anomaly interactions without level terms. Column 6: full model with levels, split anomalies, and directional interactions. The low within-R^2 (< 0.01) is standard in climate–growth panels: year-to-year weather variation accounts for only a small fraction of GDP growth variability after absorbing region and year fixed effects [Dell et al., 2012, Burke et al., 2015]. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table B2: Marginal Effects of Temperature Anomalies by Baseline Climate (Column 6)

	Scandinavia ($\bar{T} \approx 5°$C)	**Mediterranean ($\bar{T} \approx 20°$C)**
+1°C Warming	$1.20 - 0.109(5) = +0.66$ pp	$1.20 - 0.109(20) = -0.98$ pp
+1°C Cooling	$2.16 - 0.189(5) = +1.21$ pp	$2.16 - 0.189(20) = -1.62$ pp
Crossover (warming: 1.20/0.109)	$\bar{T}^* = 11.0°$C	
Crossover (cooling: 2.16/0.189)	$\bar{T}^* = 11.4°$C	

Notes: Coefficients from Column 6 of Table B1. For cold regions ($\bar{T}_i < 11°$C), *both* warming and cooling anomalies are associated with higher growth—cold acts as a structural constraint on agriculture, construction, and energy use, so any deviation is a net relaxation. For hot regions ($\bar{T}_i > 11°$C), both shock types reduce growth, indicating that economies near their thermal ceiling are fragile to *any* deviation from the optimum (heat stress from warming, disruption of heat-adapted activity from cooling).

where the cutpoints are $\{-\infty, 0, 10, 20, +\infty\}$ (in °C). We then count the number of months in each bin for each region-year:

$$B_{i,t}^{(k)} = \sum_{m=1}^{12} \mathbf{1}\{T_{i,m,t} \in [\tau_k, \tau_{k+1})\}$$

For complete years, $\sum_{k=1}^{4} B_{i,t}^{(k)} = 12$; in practice, missing monthly observations may cause the sum to fall slightly below twelve for some region-years.

C.2 Estimating Equation

The regression takes the form:

$$\ln(\text{GDP}_{i,t}) = \sum_{k=1}^{4} \beta_k \, B_{i,t}^{(k)} + \gamma \, \text{Precip}_{i,t} + \alpha_i + \delta_i \cdot t + \delta_t + \varepsilon_{i,t} \tag{11}$$

where the dependent variable is the log GDP level (scaled by 100, so coefficients approximate percentage-point differences in output); $\text{Precip}_{i,t}$ is total annual precipitation; α_i are region fixed effects; $\delta_i \cdot t$ are region-specific linear time trends; δ_t are year fixed effects; and standard errors are clustered at the NUTS-3 region level. The region-specific trends are absorbed via Frisch–Waugh–Lovell projection: within each region, all variables are regressed on an intercept and a linear time index, and the residuals are used in the main regression.

C.3 The Adding-Up Constraint and Identification

Because the bin counts approximately sum to a constant ($\sum_k B_{i,t}^{(k)} \approx 12$), the four bin variables are nearly collinear once a constant (absorbed by the region fixed effects) is included. After entity demeaning and detrending (the FWL projection), the demeaned bins sum to approximately zero for every observation. This introduces a near-perfect

linear dependence among the regressors, analogous to including a full set of indicator variables without dropping a reference category.

As a practical consequence, the *individual* coefficients β_k are identified only up to a common additive constant c: any set $\{\beta_k + c\}$ produces the same residuals and fit as $\{\beta_k\}$. However, the *differences* between any two coefficients are uniquely identified and invariant to the choice of normalization:

$$\Delta_{k,\ell} \equiv \beta_k - \beta_\ell \quad \text{is well-identified for all } k, \ell.$$

This is the standard identification result in climate econometrics when using exhaustive temperature bins [see, e.g., Dell et al., 2014]. The practical implication is that one should interpret the *shape* of the response curve—its relative ordering and curvature—rather than the absolute level of any single coefficient.

C.4 Coefficient Estimates

Table C3 reports the estimated coefficients and cluster-robust standard errors from Equation (11).

Table C3: Binned Damage Function: Estimated Coefficients

Temperature Bin	**Coefficient**	**Cluster-Robust SE**	***t*-statistic**
< 0°C	-0.49	0.05	-9.10^{***}
0–10°C	$+0.03$	0.04	$+0.81$
10–20°C	-0.03	0.03	-0.95
> 20°C	-0.19	0.06	-3.24^{**}

Notes: $^{***}p < 0.001$, $^{**}p < 0.01$. Individual bin coefficients are identified up to a common additive constant. The economically meaningful quantities are the *differences* between any two bin coefficients, as detailed below. Controls include total annual precipitation, region fixed effects, region-specific linear time trends, and year fixed effects. Standard errors are clustered at the NUTS-3 level.

C.5 Interpreting the Results: Pairwise Differences

Since only pairwise differences are identified, the correct way to read Figure 2 is to compare any two bins. Table C4 reports the key pairwise differences, using the optimal 0–10°C bin as the reference.

C.6 Interpretation

To illustrate, reallocating one month from the 0–10°C range into the hot bin (> 20°C) implies a GDP penalty of $\beta_{>20} - \beta_{0\text{-}10} = (-0.19) - (+0.03) = -0.22$ pp. Warming a cold month (< 0°C) into the 0–10°C range yields +0.52 pp, confirming the relaxation of

Table C4: Pairwise Differences Relative to the 0–10°C Bin

Reallocation	GDP Effect (pp)	Interpretation
< 0°C vs. 0–10°C	−0.52	Cold months substantially harmful
10–20°C vs. 0–10°C	−0.06	Mild months statistically indistinguishable
> 20°C vs. 0–10°C	−0.22	Hot months significantly damaging

Notes: Each entry shows the GDP impact (in percentage points) of reallocating one month from the 0–10°C range into the specified bin, after controlling for region-specific linear time trends. Bin coefficients reflect the temperature effect net of any trend-driven shifts.

cold-weather constraints. The 10–20°C bin is statistically indistinguishable from 0–10°C (−0.06 pp, $t = -0.95$), suggesting that mild and cool months are similarly productive. The qualitative shape—with 0–10°C as the optimal bin—is strikingly consistent with the parametric and GAM results. The inclusion of region-specific linear time trends attenuates the coefficient magnitudes (compared to a specification without trends), as the trends absorb low-frequency variation that is correlated with both temperature and output.

C.7 Monthly vs. Daily Bin Construction

The canonical binned estimator in climate econometrics [Deschênes and Greenstone, 2011, Dell et al., 2014] counts the number of *days* whose temperature falls into each bin. Our bins instead count *months*. This is a deliberate design choice: the appropriate temporal resolution for the temperature exposure variable should match the temporal resolution of the outcome.

Our dependent variable is annual regional GDP—an aggregate that reflects cumulative production decisions, capital allocation, and labor supply over an entire year. Year-over-year variation in annual output is likely driven more by how many months a region spent in each broad thermal regime than by whether it experienced a handful of individual extreme-temperature days. A region that endures three consecutive months above 25°C faces sustained heat stress on labor productivity, persistent cooling costs, and prolonged disruption to outdoor sectors (agriculture, construction, tourism). These cumulative, season-length exposures are precisely what monthly bins capture.

Daily bins, by contrast, are the natural choice when the dependent variable is itself a high-frequency measure—such as daily mortality [Deschênes and Greenstone, 2011], hourly labor supply [Graff Zivin and Neidell, 2014], or weekly crop growth. In those settings, acute physiological or behavioral responses to individual extreme-temperature days are the mechanism of interest, and smoothing over days within a month would attenuate the signal. For annual GDP, the relevant channel is more likely to operate at the seasonal scale, where the distribution of "cold months," "mild months," and "hot months" shapes the annual production environment.

As a cross-validation, our DML analysis (Section 4.2) uses the annual count of days with maximum temperatures exceeding 35°C as a treatment variable—a daily-resolution extreme-heat measure—and recovers the same spatial heterogeneity as the monthly bins, confirming that the qualitative pattern is robust across temporal resolutions.

D Causal Mediation Analysis: Additional Results

D.1 Results

Table D1 reports the decomposition for the hottest decile under the preferred specification, which absorbs region-specific linear time trends via Frisch–Waugh–Lovell (FWL) projection in addition to region and year fixed effects. Delta-method (Sobel) standard errors are reported for each indirect effect and cluster-robust standard errors for the total and direct effects. The total effect of a unit temperature anomaly on GDP growth is -0.582 percentage points (SE $= 0.191$, $p < 0.01$).

Labor productivity accounts for 95.9% of the total damage ($\hat{\delta}_1 = -0.558$, Sobel SE $=$ 0.124, $p < 0.001$), indicating that the productivity collapse nearly fully explains the aggregate loss. The tight 95% CI $[-0.801, -0.315]$ leaves no doubt about the economic and statistical significance of this channel. Employment, counter-intuitively, *offsets* 40.4% of the damage ($\hat{\delta}_4 = +0.235$, Sobel SE $= 0.059$, $p < 0.001$), consistent with labor hoarding or factor substitution; this buffering effect is also precisely estimated (95% CI $[+0.119$, $+0.351]$).

By contrast, investment contraction contributes 20.0% ($\hat{\delta}_2 = -0.116$, Sobel SE $=$ 0.078, $p = 0.136$) and capital stock depreciation adds 39.8% ($\hat{\delta}_3 = -0.231$, Sobel SE $=$ 0.134, $p = 0.084$). Neither channel reaches statistical significance at the 5% level, though capital stock approaches marginal significance once trends are absorbed. For investment, the wide confidence interval spans zero, reflecting uncertainty in the path coefficients rather than a near-zero estimate.

The residual direct effect is $+0.089$ (SE $= 0.076$, -15.3%), small and statistically insignificant, indicating that the four mediators collectively account for the vast majority of the total effect with little unexplained residual.

D.2 Interpretation and Caveats

The product-of-coefficients decomposition is valid under the assumption of no unobserved confounders of the mediator–outcome relationship, conditional on the treatment and controls [Imai et al., 2010]. In observational panel data, this "sequential ignorability" assumption is strong: an unobserved shock that simultaneously affects, say, labor productivity and GDP growth (beyond the temperature channel) would bias the indirect effect

Table D1: Mediation Decomposition of Temperature Effect on GDP Growth (Hottest Decile, Preferred Specification)

Channel	**Effect (pp)**	**SE**	***p*-value**	**% of Total**	**Direction**
Total Effect ($\hat{\tau}$)	−0.582	0.191	<0.01**	100.0%	Damage
Labor Productivity ($\hat{\delta}_1$)	−0.558	0.124	<0.001***	95.9%	Damage
Investment ($\hat{\delta}_2$)	−0.116	0.078	0.136	20.0%	Damage
Capital Stock ($\hat{\delta}_3$)	−0.231	0.134	0.084	39.8%	Damage
Employment ($\hat{\delta}_4$)	+0.235	0.059	<0.001***	−40.4%	Buffer
Net Direct Effect ($\hat{\delta}_0$)	+0.089	0.076	0.240	−15.3%	Residual
Fixed effects and controls					
Region FE (μ_i)	Yes (absorbed via FWL projection)				
Region-specific linear trends ($\delta_i \cdot t$)	Yes (absorbed via FWL projection)				
Year FE (θ_t)	Yes (year dummies)				
Lagged GDP growth	Yes				

Notes: Product-of-coefficients mediation decomposition. Sample restricted to regions in the hottest decile (baseline temperature ≥ 14.79°C, $N = 3{,}059$, 118 regions). Standard errors for the total and direct effects are cluster-robust at the region level; standard errors for the indirect effects are computed via the delta method (Sobel formula: $\text{SE}(\hat{a}\hat{b}) = \sqrt{\hat{a}^2\text{SE}(\hat{b})^2 + \hat{b}^2\text{SE}(\hat{a})^2}$). Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Percentages sum to approximately 100% by construction. Table D2 reports the comparison with the baseline specification without trends.

estimate. The preferred specification mitigates this concern by absorbing both time-invariant regional confounders (region FE), common aggregate shocks (year FE), and region-specific linear trends via FWL projection. Table D2 confirms that the channel structure is qualitatively robust to the inclusion of trends.

We interpret the decomposition as informative about the *relative importance* of economic channels rather than as a definitive causal partition. The delta-method standard errors (Table D1) sharpen this interpretation considerably: the labor-productivity and employment channels are estimated with sufficient precision to reject the null of zero indirect effect at the 0.1% level, whereas the investment and capital stock channels—while directionally consistent with economic priors—cannot be distinguished from zero at conventional significance levels. This pattern suggests that future work with richer microdata may profitably focus on the precisely estimated channels (productivity collapse and labor substitution) while acknowledging that the imprecisely estimated channels (investment contraction and capital depreciation) remain plausible but require additional identification or larger samples to confirm.

The near-zero and statistically insignificant residual direct effect ($p = 0.240$) provides reassurance that the four mediators collectively capture the relevant transmission mechanisms. The lack of significance indicates that the four measured channels account for the vast majority of the total temperature impact, leaving only a small, imprecisely estimated

residual for any remaining unmeasured pathways.

D.3 Sensitivity to Region-Specific Time Trends

Table D2 compares the preferred specification (with region-specific linear trends) to the baseline specification (region and year FE only, without trends) for all three shock types. This comparison quantifies the extent to which region-specific secular trends in technology, institutions, or structural transformation influence the mediation estimates.

Table D2: Mediation Sensitivity: Baseline (Without Trends) vs. Preferred (With Region-Specific Trends)

	Baseline (Region + Year FE)			**+ Region-Specific Trends**		
Channel	**Effect**	**SE**	**% of Total**	**Effect**	**SE**	**% of Total**
Panel A: Temperature (Hottest Decile)						
Total Effect ($\hat{\tau}$)	-0.867^{***}	0.178	100.0%	-0.582^{**}	0.191	100.0%
Labor Productivity ($\hat{\delta}_1$)	-0.717^{***}	0.121	82.7%	-0.558^{***}	0.124	95.9%
Investment ($\hat{\delta}_2$)	-0.175^{*}	0.075	20.2%	-0.116	0.078	20.0%
Capital Stock ($\hat{\delta}_3$)	-0.150	0.128	17.4%	-0.231	0.134	39.8%
Employment ($\hat{\delta}_4$)	$+0.219^{***}$	0.056	-25.3%	$+0.235^{***}$	0.059	-40.4%
Direct Effect ($\hat{\delta}_0$)	-0.044	0.068	5.1%	$+0.089$	0.076	-15.3%
Panel B: Flood Anomaly (Full Sample)						
Total Effect ($\hat{\tau}$)	-0.031^{***}	0.003	100.0%	-0.026^{***}	0.004	100.0%
Labor Productivity ($\hat{\delta}_1$)	-0.020^{***}	0.003	62.6%	-0.016^{***}	0.003	61.6%
Investment ($\hat{\delta}_2$)	-0.010^{***}	0.001	33.6%	-0.010^{***}	0.001	38.6%
Capital Stock ($\hat{\delta}_3$)	-0.008^{***}	0.002	26.7%	-0.005^{*}	0.002	17.6%
Employment ($\hat{\delta}_4$)	-0.001	0.001	4.1%	-0.002	0.001	5.9%
Direct Effect ($\hat{\delta}_0$)	$+0.008^{***}$	0.001	-27.0%	$+0.006^{**}$	0.001	-23.7%
Panel C: Drought Anomaly (Full Sample)						
Total Effect ($\hat{\tau}$)	-0.007	0.006	100.0%	-0.012^{*}	0.006	100.0%
Labor Productivity ($\hat{\delta}_1$)	-0.002	0.004	26.5%	-0.003	0.004	28.3%
Investment ($\hat{\delta}_2$)	-0.002	0.001	28.4%	-0.003^{*}	0.001	23.6%
Capital Stock ($\hat{\delta}_3$)	$+0.002$	0.003	-27.1%	-0.002	0.003	16.4%
Employment ($\hat{\delta}_4$)	-0.004^{**}	0.001	55.6%	-0.006^{***}	0.001	47.7%
Direct Effect ($\hat{\delta}_0$)	-0.001	0.002	16.5%	$+0.002^{*}$	0.002	-16.0%

Notes: Product-of-coefficients mediation decomposition. Panel A restricts to the hottest decile (baseline temperature ≥ 14.79°C); Panels B–C use the full sample with temperature controls (matching Model 6). "Baseline" includes: region FE (within transformation) + year FE (year dummies) + lagged GDP growth. "+ Region-Specific Trends" additionally absorbs region-specific linear time trends ($\delta_i \cdot t$) via Frisch–Waugh–Lovell projection. Neither specification includes country×year FE (see Table D3 for that comparison). Standard errors: cluster-robust at the region level (total and direct effects) or delta method/Sobel (indirect effects). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Three patterns emerge from the comparison. First, the temperature total effect is attenuated by approximately one-third when region-specific trends are absorbed (-0.87

to −0.58 pp), consistent with trends capturing slow-moving structural changes that covary with both temperature and growth. However, the effect remains statistically significant at the 1% level, confirming that the temperature–growth relationship is not an artifact of unmeasured linear confounders.

Second, the *channel decomposition is qualitatively stable*: labor productivity remains the dominant channel (83% → 96% of the total effect), employment continues to buffer damages (−25% → −40%), and investment remains a modest damage channel. The increase in the labor-productivity share from 83% to 96% reflects the attenuation of the total effect (denominator) more than a change in the indirect effect (numerator), indicating that the productivity channel is particularly robust to the inclusion of trends.

Third, the precipitation results are largely unchanged: the flood channel structure is stable across specifications (productivity: 63% → 62%; investment: 34% → 39%), and the drought total effect becomes marginally significant ($p = 0.03$) once trends are absorbed, suggesting that low-frequency precipitation trends may have masked some drought damage in the baseline specification.

D.4 Sensitivity to Country-Year Fixed Effects

The preferred mediation specification absorbs common time effects via year fixed effects, identifying the temperature–mediator–GDP relationship from within-region variation after netting out Europe-wide shocks. An alternative approach replaces year fixed effects with country×year fixed effects, which absorb *national* business cycles, fiscal policy changes, and country-wide climate anomalies. This more restrictive specification identifies the effect exclusively from *within-country geographic* variation in temperature exposure.

Table D3 reports the comparison for the hottest decile. Three patterns emerge.

First, the *total effect strengthens* from −0.582 pp to −0.897 pp ($p < 0.001$) under country×year fixed effects. This 54% increase suggests that national macroeconomic dynamics—such as tourism-driven expansions in hot Mediterranean years—partially offset the estimated temperature damage under year-only fixed effects, and that absorbing these dynamics reveals a larger underlying penalty.

Second, *when controlling for national business cycles (country×year fixed effects), the employment buffer completely dissipates.* The employment indirect effect changes from +0.235 pp ($p < 0.001$) and reverses sign to an insignificant −0.091 pp ($p = 0.380$). This indicates that labor buffering is not predominantly a localized, within-country adaptation by highly constrained individual firms. Rather, it is driven by cross-country differences in labor-market rigidities and national-scale sectoral shifts (e.g., pan-Mediterranean tourism labor flows during hot years). By absorbing these national-level dynamics, the country×year fixed effects reveal that the employment buffer is a macro-regional phe-

Table D3: Mediation Sensitivity: Year FE vs. Country×Year Fixed Effects (Hottest Decile)

	Preferred (Year FE + Trends)			**Country×Year FE**		
Channel	**Effect**	**SE**	**% of Total**	**Effect**	**SE**	**% of Total**
Total Effect ($\hat{\tau}$)	-0.582^{**}	0.191	100.0%	-0.897^{***}	0.263	100.0%
Labor Productivity ($\hat{\delta}_1$)	-0.558^{***}	0.124	95.9%	-0.388	0.260	43.3%
Investment ($\hat{\delta}_2$)	-0.116	0.078	20.0%	-0.041	0.081	4.6%
Capital Stock ($\hat{\delta}_3$)	-0.231	0.134	39.8%	-0.219	0.225	24.4%
Employment ($\hat{\delta}_4$)	$+0.235^{***}$	0.059	-40.4%	-0.091	0.104	10.1%
Direct Effect ($\hat{\delta}_0$)	$+0.089$	0.076	-15.3%	-0.158	—	17.6%
Fixed effects and controls						
Region FE (μ_i)	Yes (FWL)			Yes (within)		
Year FE (θ_t)	Yes (dummies)			—		
Region trends ($\delta_i \cdot t$)	Yes (FWL)			—		
Country×Year FE	—			Yes (within)		
Lagged GDP growth	Yes			Yes		

Notes: Product-of-coefficients mediation for the hottest decile (baseline temperature ≥ 14.79°C, $N = 3{,}059$, 118 regions across 8 countries: CY, EL, ES, FR, HR, IT, MT, PT). "Preferred" absorbs region FE and region-specific linear trends via FWL projection, plus year dummies. "Country×Year FE" absorbs region FE and country×year FE via sequential within-group demeaning (year dummies are redundant and excluded). Standard errors: cluster-robust at the region level (total and direct effects) or delta method/Sobel (indirect effects). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

nomenon. The effect is also purely contemporaneous: at a one-year lag, the employment channel vanishes (Path A = +0.04, $p = 0.80$), suggesting that any factor substitution does not persist beyond the year of the shock.

Third, the *labor-productivity share declines* from 96% to 43% of the larger total effect, with a corresponding increase in the direct (residual) effect. This redistribution of explanatory shares, combined with the inflated standard errors (the 8-country subset struggles to support channel-level inference with 208 additional fixed effects), highlights the fragility of decomposing overlapping channels when the primary variation is soaked up.

In summary, the qualitative conclusion—that temperature damages in hot European regions are driven primarily by productivity collapse—is robust to region-specific linear time trends (Table D2) and survives the country×year specification at the total-effect level. The employment buffer, however, is a national macroeconomic response rather than a structural micro-level adaptation, and requires identifying variation across country borders to estimate.

E Sectoral Temperature Sensitivity: Methodology and Findings

E.1 Motivation and Research Question

The main analysis reveals that temperature shocks reduce regional GDP growth, with particular severity in hot Mediterranean regions. However, the aggregate GDP loss conceals heterogeneous sectoral responses. A critical follow-on question is: *which sectors drive the overall economic damage?* Understanding sectoral vulnerability is essential for designing targeted adaptation policy. Does agricultural collapse dominate, as historical patterns suggest? Are modern service economies insulated? How do construction and manufacturing adapt?

To address this question, we extend the panel fixed-effects framework to sector-level value-added data for four broad economic sectors: Agriculture, Manufacturing, Construction, and Services. We employ a comparative research design, analyzing temperature impacts separately for the hottest and coldest European regions (decile samples). This design reveals whether sector vulnerabilities are universal or regime-specific, illuminating fundamental differences in climate adaptation constraints across the thermal gradient.

E.2 Data: Hottest and Coldest Regions

We focus on two contrasting regional subsets:
Hottest Regions (90th percentile baseline temperature):

- **Temperature threshold:** ≥ 14.79°C baseline annual mean
- **Number of regions:** 118 regions (10.1% of sample)
- **Geographic concentration:** Cyprus (CY000, 20.16°C), southern Spain (ES640, 19.50°C), Malta (MT001–MT002, 19.3°C), Greece (EL421–EL432, 18.8–19.2°C), southern Italy (ITG11, 18.39°C)
- **Observations:** 3,049 region-year observations per sector (1991-2020 baseline, full 2000-2020 analysis period)
- **Regional count for regressions:** 118 regions (cluster unit for standard errors)

Coldest Regions (10th percentile baseline temperature):

- **Temperature threshold:** ≤ 8.10°C baseline annual mean
- **Number of regions:** 117 regions (10.0% of sample)

- **Geographic concentration:** Alpine Austria (AT333, 1.32°C), Finnish Lapland (FI1D7, FI1D8, FI1D9, FI1D3: 1.35–3.53°C), Swedish inland (SE331, SE332, SE322: 1.90–3.21°C), Northern Italian Alps (ITC20, 2.84°C)
- **Observations:** 3,084 region-year observations per sector
- **Regional count for regressions:** 117 regions

Sectoral value-added data are sourced from the national accounts GVA NACE compilation, which breaks regional gross value added into four broad classifications aligned with the NACE hierarchy. Agriculture includes crop and livestock production; Manufacturing includes all secondary production; Construction includes real estate and building services; and Services captures everything else (finance, trade, transport, government, health, education, and culture).

E.3 Estimating Equation

The sectoral temperature response is estimated using a panel fixed-effects model with demeaning:

$$\Delta \ln S_{i,t} = \beta_1^+ \Delta T_{i,t}^+ + \beta_2^+ (\Delta T_{i,t}^+)^2 + \beta_1^- \Delta T_{i,t}^- + \beta_2^- (\Delta T_{i,t}^-)^2 + \beta_3 \ln S_{i,t-1} + \beta_4 \ln S_{i,t-2} + \alpha_i + \gamma_t + \epsilon_{i,t} \quad (12)$$

where:

- $\Delta \ln S_{i,t} = \ln S_{i,t} - \ln S_{i,t-1}$ is the growth rate (log-difference) of sectoral value-added in region i during year t
- $\Delta T_{i,t}$ is the annual temperature anomaly (observed temperature minus 1991–2020 climatological baseline for that region)
- $\Delta T_{i,t}^+ = \max(\Delta T_{i,t}, 0)$ is the warming anomaly, allowing separate estimation of warming impacts
- $\Delta T_{i,t}^- = \min(\Delta T_{i,t}, 0)$ is the cooling anomaly, allowing separate estimation of cooling impacts
- $(\Delta T_{i,t}^+)^2$ and $(\Delta T_{i,t}^-)^2$ capture non-linear responses within warming and cooling regimes
- $\ln S_{i,t-1}$ and $\ln S_{i,t-2}$ are lagged log sector sizes, controlling for mean reversion and recent sectoral trends
- α_i is the region fixed effect (absorbs all permanent regional characteristics: geography, infrastructure, industrial structure, institutions)

- γ_t is the year fixed effect (removes aggregate macroeconomic cycles, technological trends, EU policies)
- $\epsilon_{i,t}$ is the idiosyncratic error term

E.3.1 Estimation via Within Transformation

The model is estimated using the *within transformation* (demeaning), a standard approach for two-way fixed-effects panel models:

1. **Demean within regions:** For each region i, compute $\tilde{X}_{i,t} = X_{i,t} - \bar{X}_i$ where $\bar{X}_i$ is the time-mean within that region across all years. This eliminates the region fixed effect α_i.
2. **Demean within years:** For each year t, compute $\ddot{X}_{i,t} = \tilde{X}_{i,t} - \bar{\tilde{X}}_t$ where $\bar{\tilde{X}}_t$ is the cross-sectional mean of demeaned variable $\tilde{X}$ in year t. This eliminates the year fixed effect γ_t.
3. **Estimate by OLS:** Run OLS regression of the twice-demeaned growth rate on twice-demeaned regressors:

$$\Delta \ddot{\ln} S_{i,t} = \beta_1^+ \Delta\ddot{T}_{i,t}^+ + \beta_2^+ (\Delta\ddot{T}_{i,t}^+)^2 + \beta_1^- \Delta\ddot{T}_{i,t}^- + \beta_2^- (\Delta\ddot{T}_{i,t}^-)^2 + \beta_3 \ddot{\ln} S_{i,t-1} + \beta_4 \ddot{\ln} S_{i,t-2} + \ddot{\epsilon}_{i,t}$$

The demeaning procedure completely eliminates both time-invariant regional heterogeneity and common time trends, leaving only within-region, within-year variation to identify the temperature–growth relationship. This is critical because regions differ in permanent characteristics (soil quality, industrial composition, latitude) that correlate with both baseline temperature and sectoral productivity.

E.3.2 Standard Errors and Inference

Standard errors are estimated using a *Conley spatial-HAC estimator* with a Bartlett kernel and a 500 km distance cutoff, applied to region-centroid geodesic distances. As discussed in Section 3.3, this estimator directly models the spatial and temporal decay of error covariances, subsuming the within-region cluster correction (since within-region distance is zero, all within-region correlations receive full kernel weight). The resulting variance estimator replaces the standard block-diagonal cluster meat matrix with a full spatial covariance matrix:

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{S}'\mathbf{W}\mathbf{S})(\mathbf{X}'\mathbf{X})^{-1} \times \mathrm{adj}$$

where $\mathbf{S}$ is the matrix of region-level score vectors, $\mathbf{W}$ is the spatial weight matrix using a Bartlett kernel thresholded at 500 km, and adj is the degree-of-freedom adjust-

ment. This allows for arbitrary serial correlation within regions over time, as well as cross-sectional spatial correlation across neighboring regions up to 500 km away (e.g., contiguous heatwaves or unmeasured regional technological spillovers).

E.4 Results for Hottest Regions

Table E1 summarizes sectoral temperature sensitivity in the 118 hottest European regions.

Table E1: **Sectoral Temperature Impacts: Hottest Regions (≥ 14.79°C baseline)**

	Warming Coefficients		Cooling Coefficients		Growth Impact	
Sector	β_1^+ (SE)	β_2^+ (SE)	β_1^- (SE)	β_2^- (SE)	@ +2°C	@ −2°C
Agriculture	+1.873 (3.031)	−3.813 (2.321)	−7.351* (3.875)	−6.163 (3.965)	−11.505*** pp	−9.949 pp
Manufacturing	+1.201 (1.638)	−1.058 (1.260)	−0.266 (3.007)	−0.150 (3.103)	−1.829 pp	−0.069 pp
Construction	+8.026*** (1.922)	−2.299 (1.581)	−21.071*** (3.934)	−15.329*** (4.600)	+6.857** pp	−19.174* pp
Services	+0.998* (0.583)	−1.089** (0.484)	−2.414** (0.964)	−1.793 (1.115)	−2.358** pp	−2.346 pp

Notes: All regressions include region FE and year FE (two-way panel FE). Warming and cooling anomalies are modeled separately with linear and quadratic terms. Impact significance is computed via delta method using the estimated variance-covariance matrix. SE are cluster-robust (clustered at the NUTS-3 region level). Significance levels: $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E.4.1 Agriculture: Severe Vulnerability

Agriculture in hot regions shows asymmetric responses to warming and cooling anomalies rather than a single symmetric effect.

With anomalies split by sign, the warming linear term is positive but imprecise ($\beta_1^+ = +1.87$, p = 0.54) and the warming quadratic term is negative ($\beta_2^+ = -3.81$, p = 0.10). The cooling linear term is negative and marginally significant ($\beta_1^- = -7.35^*$, p = 0.058) with an imprecise cooling quadratic term ($\beta_2^- = -6.16$, p = 0.12). At +1°C warming, the estimated impact is −1.94 pp (p = 0.13), while at −1°C cooling the impact is +1.19 pp (p = 0.59).

These results suggest modest asymmetry near the baseline, with damages emerging more clearly at larger positive anomalies. The mechanism remains straightforward: heat stress reduces crop yields, forage quality for livestock, and farm labor productivity. Mediterranean regions that depend on irrigation-intensive crops (fruits, vegetables, wine) face compound stress from both direct heat damage and water scarcity.

E.4.2 Construction: Paradoxical Non-Monotonicity

The construction sector's response is highly non-linear and counterintuitive. The warming linear coefficient is strongly positive ($\beta_1^+ = +8.03^{***}$, p ¡ 0.001) with an imprecise warming quadratic term ($\beta_2^+ = -2.30$, p = 0.15). On the cooling side, both the linear and

quadratic terms are strongly negative ($\beta_1^- = -21.07^{***}$, $\beta_2^- = -15.33^{***}$), implying sharp non-linearity for cooling anomalies.

At +1°C warming, the estimated impact is +5.73 pp (p ¡ 0.01). At −1°C cooling, the estimated impact is +5.74 pp (p = 0.001). This pattern suggests that even modest anomalies trigger a surge in construction activity—likely reflecting rapid post-damage reconstruction. Heat waves that damage buildings, infrastructure, and agricultural facilities trigger emergency building, repairs, and replacement. Alternatively, anticipatory construction may increase as regions attempt to upgrade infrastructure (air conditioning, cooling systems, flood barriers) in response to perceived climate risks.

E.4.3 Manufacturing: No Significant Effect

Manufacturing displays no statistically significant warming or cooling coefficients in hot regions. Both the warming terms ($\beta_1^+ = +1.20$, $\beta_2^+ = -1.06$) and cooling terms ($\beta_1^- = -0.27$, $\beta_2^- = -0.15$) are imprecise, and the impacts at ±1°C are near zero. This suggests that modern manufacturing in hot European regions either (i) operates in climate-controlled indoor environments insulated from ambient heat; (ii) has already adapted via technological investment; or (iii) is less exposed to temperature variability than outdoor-dependent agriculture. This finding aligns with micro-economic evidence showing manufacturing is less thermally sensitive than agriculture in firm-level European data [Gagliardi et al., 2024].

E.4.4 Services: Modest Vulnerability

Services show asymmetric responses, with a marginally positive warming linear term ($\beta_1^+ = +1.00^{*}$) and a significant negative warming quadratic term ($\beta_2^+ = -1.09^{**}$). Cooling anomalies are also damaging in the linear term ($\beta_1^- = -2.41^{**}$), while the cooling quadratic term is imprecise. The estimated impact at +1°C is −0.09 pp (ns), and at −1°C it is +0.62 pp (ns), indicating modest effects around the baseline. These patterns suggest either direct heat stress on service workers (particularly those in low-paid, outdoor-exposed service occupations like tourism, hospitality, retail) or indirect demand destruction from heat-induced household budget constraints. However, the effect is smaller than agriculture, consistent with services being a "resilience sector" in hot climates.

E.5 Results for Coldest Regions

We now turn to the same sectoral analysis in the 117 coldest European regions (Table E2). The contrast with hot regions is stark and illuminating.

Table E2: **Sectoral Temperature Impacts: Coldest Regions (≤ 8.10°C baseline)**

	Warming Coefficients		Cooling Coefficients		Growth Impact	
Sector	β_1^+ (SE)	β_2^+ (SE)	β_1^- (SE)	β_2^- (SE)	@ +2°C	@ −2°C
Agriculture	−2.398 (2.196)	+2.034* (1.177)	−2.495 (2.067)	+1.665 (1.019)	+3.343* pp	+11.649*** pp
Manufacturing	−0.926 (1.370)	+0.741 (0.640)	+4.857*** (1.265)	+4.308*** (0.862)	+1.110 pp	+7.518*** pp
Construction	−1.760 (1.978)	+0.817 (0.804)	+2.405 (1.498)	+1.561* (0.881)	−0.253 pp	+1.432 pp
Services	−0.235 (0.497)	+0.399 (0.257)	+1.325* (0.710)	+0.795** (0.345)	+1.127** pp	+0.530 pp

Notes: Significance levels: $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Warming and cooling anomalies are modeled separately with linear and quadratic terms. Impact significance is computed via delta method using the estimated variance-covariance matrix. SE are cluster-robust (clustered by region).

E.5.1 Agriculture: Warming Benefits Offset Marginal Cooling Losses

Cold-region agriculture shows asymmetric warming and cooling responses. The warming linear term is negative but imprecise ($\beta_1^+ = -2.40$, p = 0.28) with a marginally positive warming quadratic term ($\beta_2^+ = +2.03^*$, p = 0.084). The cooling terms are negative and imprecise in the linear component ($\beta_1^- = -2.50$, p = 0.23) with a positive but imprecise quadratic term ($\beta_2^- = +1.66$, p = 0.10). At +1°C warming, the estimated impact is −0.36 pp (ns). At −1°C cooling, the impact is +4.16 pp (p = 0.001).

The economic interpretation reveals important asymmetry from hot regions: while hot-region agriculture shows modest damages at +1°C warming (−1.94 pp, ns), cold-region agriculture exhibits sizable gains under cooling shocks at −1°C. This reflects fundamental climate physiology. Cold-region agriculture (grains, potatoes, hardy vegetables) faces a biological constraint: short growing seasons. Cooling shocks compress the frost-free season and can trigger large yield losses or gains depending on the anomaly sign, while mild warming generates smaller net changes near baseline.

E.5.2 Manufacturing: Quadratic Benefits from Moderate Warming

Cold-region manufacturing shows no significant warming terms ($\beta_1^+ = -0.93$, $\beta_2^+ = +0.74$, both ns), while cooling terms are strongly positive and significant ($\beta_1^- = +4.86^{***}$, $\beta_2^- = +4.31^{***}$). Despite this, the impact at −1°C is −0.55 pp (ns) because the negative anomaly interacts with the positive linear term, and benefits only emerge at larger cooling magnitudes as the quadratic term dominates. At +1°C warming, the impact is −0.19 pp (ns).

This implies that cold-region manufacturing is resilient to small anomalies around the baseline, with nonlinear gains or losses emerging only under larger deviations. The mechanism is consistent with engineering constraints: moderate variation is manageable, while larger excursions (whether cold or warm) exceed the design envelope of existing infrastructure.

E.5.3 Construction and Services: Muted Responses

Construction and Services show muted responses in cold regions. Construction has no significant warming terms and only a marginal cooling quadratic term ($\beta_2^- = +1.56^*$), yet the impacts at ± 1°C remain insignificant. Services show no significant warming terms, while cooling terms are positive and significant ($\beta_1^- = +1.33^*$, $\beta_2^- = +0.79^{**}$), but the -1°C impact remains small and insignificant. Overall, temperature does not significantly constrain construction and services in already-cold regions at modest anomaly sizes, in sharp contrast to hot regions where construction and services respond strongly.

E.6 Comparative Interpretation

E.6.1 Asymmetric Climate Vulnerability

The contrasting sectoral responses in hottest versus coldest regions reveal a fundamental economic principle: *climate damages are not symmetric around an optimum temperature.* Cold-region economies experience relative benefits from modest warming because they operate far below historical biological and engineering optima. Hot-region economies, operating near or above optima, experience damages from further warming because their biological and technological systems are already thermally constrained.

Agriculture best illustrates this asymmetry. In cold regions, agriculture shows little response to +1°C warming (-0.36 pp, ns) but large gains under -1°C cooling ($+4.16^{**}$ pp), reflecting strong non-linearity in the cooling regime. In hot regions, the +1°C impact is negative (-1.94 pp, ns), with larger damages emerging at higher warming anomalies. The inflection point (where marginal benefits switch to marginal damages) occurs around 11–13°C, precisely the observed optimal temperature in the main analysis. This sectoral heterogeneity is the mechanism by which aggregate regional GDP displays a concave relationship with temperature, peaking near 11°C.

E.6.2 Construction as a Damage Indicator

The *construction sector* serves as a "canary in the coal mine" for climate damages. In hot regions, warming shocks have a strong positive linear response ($\beta_1^+ = +8.03^{***}$), while cooling terms are strongly negative ($\beta_1^- = -21.07^{***}$, $\beta_2^- = -15.33^{***}$). Yet the estimated impacts at ± 1°C are both positive and significant (around +5.7 pp), consistent with reconstruction surges following shocks. In cold regions, construction responses remain muted at ± 1°C. This suggests that climate damages in hot regions, while not directly reducing construction output, instead redirect labor and capital into reconstruction, signaling that underlying damage is economically significant. The surge in construction growth following heat shocks is not beneficial in a welfare sense—it reflects costly adaptation to damage rather than genuine productivity growth.

E.6.3 Manufacturing Resilience and Engineering Optima

Manufacturing is the most thermally resilient sector across both hot and cold regions. In hot regions, neither warming nor cooling coefficients are significant, and impacts at ± 1°C are near zero. In cold regions, warming terms remain insignificant while cooling terms are strongly positive, yet the -1°C impact is still small and insignificant, indicating that benefits emerge only at larger cooling anomalies. This reflects modern manufacturing's dependence on climate-controlled indoor environments and global supply chains less sensitive to local temperature than agriculture.

E.6.4 Services: Labor Productivity and Demand Channels

Services show asymmetric responses in hot regions (significant negative cooling linear term and negative warming curvature), while cold regions show significant cooling terms but negligible impacts at ± 1°C. This pattern suggests two mechanisms:

1. **Direct labor productivity:** Service workers in hot regions (hospitality, retail, transportation, outdoor service occupations) experience heat stress, reducing effort and output. Service workers in cold regions may initially benefit from warming if labor productivity improves. However, we observe no significant effect in cold regions, possibly because baseline cold is not extreme enough to substantially impair service work (unlike agricultural labor, which has hard biological constraints).

2. **Demand destruction:** Heat-stressed households and firms in hot regions may reduce service consumption (entertainment, dining, services require discretionary spending). Cold-region demand for services is not boosted by modest warming, likely because even at baseline temperatures, cold regions are not so thermally stressed that marginal warming noticeably increases demand for cooling, recreation, or other services.

E.7 Economic Intuition

The heterogeneous sectoral responses reflect fundamental differences in how production technologies interact with thermal environments. Agriculture is directly constrained by crop physiology (hard thresholds for germination, pollination, and photosynthesis), making it most vulnerable in hot regions and a net beneficiary of warming in cold regions. Manufacturing operates in climate-controlled indoor environments where HVAC absorbs ambient variation, explaining its resilience across both climate regimes. Construction's non-monotonic response in hot regions reflects post-damage reconstruction demand rather than genuine productivity gains—a surge triggered by infrastructure repair, not economic expansion. Services combine climate-insulated high-end occupations (finance, education)

with thermally exposed low-end occupations (hospitality, outdoor retail, care work), producing modest aggregate sensitivity dominated by labor-productivity effects in hot regions.

E.8 Implications for the Main Analysis

The sectoral composition of temperature damages has important policy implications:

1. **Agriculture-driven aggregate effect:** The agricultural sector still drives aggregate vulnerability, even under the split-anomaly specification. In hot regions, the +1°C impact is −1.94 pp (ns), and damages grow at larger positive anomalies. Agricultural value-added represents 5–20% of Mediterranean regional GDP, so a 1.94 pp sectoral shock translates to a 0.10–0.39 pp aggregate shock, consistent with observed aggregate elasticities.

2. **Reconstruction is not growth:** The construction sector's positive response in hot regions is a diagnosis of underlying damage, not a net positive. While it contributes to measured GDP, it reflects costly adaptation to climate damage, not welfare improvement.

3. **Manufacturing resilience:** The lack of manufacturing damage in hot regions, and modest benefits in cold regions, suggests that modern industrial policy can viably adapt. However, this resilience requires capital investment (HVAC, resilient supply chains) that poorer regions may struggle to afford, potentially exacerbating spatial inequality.

4. **Services vulnerability in hot regions:** The service-sector impact at +1°C is small and statistically insignificant (−0.09 pp), but the negative warming curvature suggests losses accelerate with larger warming anomalies. If extreme heat events (¿35°C) become more frequent, outdoor-dependent services (tourism, agriculture-linked services, construction) face accelerating losses with non-linear threshold effects not captured in a linear specification.

5. **Cold regions' near-term adaptation advantage:** Cold regions experience relative growth benefits from warming (especially in agriculture), suggesting that near-term (to 2050) climate change may support growth in Nordic and Alpine regions. However, if warming continues beyond 2°C per region, these regions will eventually cross their own thermal optima, shifting from adaptation advantage to vulnerability. This implies that long-term (post-2050) climate policy should focus not just on emissions mitigation, but on helping cold regions prepare for future crossing of their own productivity optima.

F Precipitation Anomalies: Transmission and Sectoral Impacts

F.1 Overview

This section extends the mediation and sectoral analysis framework from Appendices D and E to precipitation, examining how hydrological shocks transmit through the economy and which sectors face the greatest vulnerability.

Variable construction. For each region i, calendar month m, and year t, the Copernicus ERA5 reanalysis provides the deviation of monthly total precipitation from the region-specific 1991–2020 climatological mean, $\Delta P_{i,m,t}$ (in mm). We aggregate to the annual level by averaging over the twelve calendar months:

$$P_{i,t}^{\text{anom}} \;=\; \frac{1}{12}\sum_{m=1}^{12} \Delta P_{i,m,t},$$

and decompose into a flood component, $\text{flood}_{i,t} = \max\big(P_{i,t}^{\text{anom}},\ 0\big)$, and a drought component, $\text{drought}_{i,t} \;=\; \big|\min\big(P_{i,t}^{\text{anom}},\ 0\big)\big|$. In the estimation sample, $P_{i,t}^{\text{anom}}$ ranges from approximately -78 to $+84$ mm (IQR: -7 to $+7$ mm). All precipitation coefficients are per mm of average monthly anomaly.

F.2 Parametric Evidence on Precipitation Effects

Our parametric fixed-effects models (Table B1) reveal that precipitation anomalies exert economically and statistically significant negative effects on regional GDP growth:

Table F1: Precipitation Effects on Regional GDP Growth: Parametric Evidence

Precipitation Variable	Coefficient	Std. Error	P-value
Precip Anom Flood (per mm)	-0.0289***	0.0034	< 0.001
Precip Anom Drought (per mm)	-0.0116*	0.0052	0.027

Notes: Results from Table A (Appendix A, full parametric specification with directional temperature interactions, Model 6). This specification absorbs region and year fixed effects, controls for temperature levels (T, T^2), directional temperature–baseline interactions, and clusters standard errors at the region level. Sample: 31,622 region-year observations. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

These effects translate to economically significant damages. A drought year at the 5th percentile of the anomaly distribution (average monthly deficit of $\sim$18 mm) reduces annual GDP growth by approximately 0.2 percentage points (0.0116×18), while a flood year at the 95th percentile ($\sim$18 mm monthly excess) reduces growth by approximately

0.5 percentage points (0.0289×18). At the observed extremes ($\sim$80 mm), implied damages reach 1–2 percentage points. The asymmetry—floods more damaging per mm than droughts—likely reflects that most European regions have baseline precipitation near optimal levels; precipitation excess creates immediate inundation damage, whereas deficits operate through slower soil-moisture depletion channels.

F.3 Causal Mediation: Precipitation Transmission Channels

We now decompose the total effect of precipitation shocks on GDP growth into contributions from four economic channels, using the product-of-coefficients approach [Baron and Kenny, 1986] identical to the temperature mediation in Appendix D.

Specification. The mediation regressions replicate the control structure of the parametric Model 6 (Appendix B, Table B1) that generates the Table E1 coefficients, adding only lagged GDP growth as a convergence control. Denoting the treatment of interest by $P_{i,t}$ (either $\text{flood}_{i,t}$ or $\text{drought}_{i,t}$), the other precipitation anomaly by $P^{-}_{i,t}$, and the vector of temperature controls by $\mathbf{Z}_{i,t} = (T_{i,t},\ T^2_{i,t},\ W_{i,t}\times\bar{T}_i,\ C_{i,t}\times\bar{T}_i)$, the three-equation system for each mediator $M_{k,i,t}$ is:

Table E1 (parametric Model 6, for reference):

$$g_{i,t} \;=\; \beta_1 \,\text{flood}_{i,t} + \beta_2 \,\text{drought}_{i,t} + \mathbf{Z}'_{i,t}\boldsymbol{\gamma} + \mu_i + \theta_t + \varepsilon_{i,t} \tag{13}$$

Table E2/E3 mediation — Total effect:

$$g_{i,t} \;=\; \tau\, P_{i,t} + \alpha\, P^{-}_{i,t} + \mathbf{Z}'_{i,t}\boldsymbol{\gamma} + \phi\, g_{i,t-1} + \mu_i + \theta_t + u_{i,t} \tag{14}$$

Path A (treatment → mediator):

$$M_{k,i,t} \;=\; a_k\, P_{i,t} + \alpha^A_k\, P^{-}_{i,t} + \mathbf{Z}'_{i,t}\boldsymbol{\gamma}^A_k + \phi^A_k\, g_{i,t-1} + \mu^A_i + \theta^A_t + u^A_{k,i,t} \tag{15}$$

Path B (mediator → GDP, controlling for treatment):

$$g_{i,t} \;=\; \tau'_k\, P_{i,t} + b_k\, M_{k,i,t} + \alpha^B_k\, P^{-}_{i,t} + \mathbf{Z}'_{i,t}\boldsymbol{\gamma}^B_k + \phi^B_k\, g_{i,t-1} + \mu^B_i + \theta^B_t + u^B_{k,i,t} \tag{16}$$

The indirect effect of channel k is $\hat{\delta}_k = \hat{a}_k \times \hat{b}_k$, with Sobel standard errors; the percentage contribution is $100 \times \hat{\delta}_k/\hat{\tau}$; and the residual direct effect is $\hat{\delta}_0 = \hat{\tau} - \sum_k \hat{\delta}_k$.

Equations (13) and (14) share the same controls ($\mathbf{Z}_{i,t}$, $P^{-}_{i,t}$, μ_i, θ_t), so $\hat{\tau}$ in the mediation closely matches $\hat{\beta}_1$ or $\hat{\beta}_2$ from Table E1. The mediation sample (30,476 observations, 1,146 regions) is slightly smaller than the parametric sample (31,622) because equation (14)

additionally includes $g_{i,t-1}$, which drops the first observed year. Standard errors are cluster-robust at the region level throughout.

F.3.1 Flood Anomaly Transmission

Table F2: Mediation Decomposition: Flood Anomaly → GDP Growth

Channel	**Path A ($\hat{a}$)**	**Path B ($\hat{b}$)**	**Indirect ($\hat{\delta}$)**	**% of Total**
Total Effect ($\hat{\tau}$)			-0.0311^{***}	100.0%
Labor Productivity	-0.0279^{***}	$+0.7999^{***}$	-0.0223	71.8%
Investment (GFCF)	-0.0777^{***}	$+0.1344^{***}$	-0.0104	33.6%
Capital Stock	-0.0106^{***}	$+0.7861^{***}$	-0.0083	26.7%
Employment	-0.0031	$+0.4122^{***}$	-0.0013	4.1%
Direct Effect ($\hat{\delta}_0$)			$+0.0113$	-36.2%

Notes: Product-of-coefficients decomposition. All regressions include region FE (within transformation), year FE (year dummies), temperature controls (matching Model 6), both precipitation anomalies, and lagged GDP growth. Region-specific trends are *not* included (baseline specification). Cluster-robust SEs at the NUTS-3 region level. $N = 30{,}476$, 1,146 regions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Percentages: positive = damage amplifying channel; negative = buffer.

The total effect of flood anomalies on GDP growth is -0.0311 pp per mm of excess precipitation ($p < 0.001$), consistent with the parametric estimate of -0.0289 in Table E1. The decomposition reveals:

- **Labor productivity (71.8%):** The dominant transmission channel. Floods reduce output per hour worked through work disruptions, damaged equipment, contaminated work environments, and reduced efficiency during cleanup. Path A ($\hat{a} = -0.028^{***}$) confirms that floods directly depress productivity, and Path B ($\hat{b} = +0.800^{***}$) confirms productivity's strong link to GDP.
- **Capital stock destruction (26.7%):** Physical capital destruction (buildings, machinery, infrastructure) transmits 26.7% of the total flood damage. Path A ($\hat{a} = -0.011^{***}$) indicates capital stock declines significantly following floods.
- **Investment contraction (33.6%):** Gross fixed capital formation declines ($\hat{a} = -0.078^{***}$), likely as firms defer planned investment during and after flood events. This path accounts for a third of total damages.
- **Employment (4.1%):** Employment shows a small, insignificant decline ($\hat{a} = -0.003$), contributing modestly to the total damage. Unlike temperature shocks—where employment buffers 40.4% of damage—flood shocks leave little room for labor-market adjustment.
- **Direct effect (-36.2%):** The positive residual suggests the four mediators collectively *overestimate* the total effect mildly, implying either partial offsets through unmeasured channels (e.g., post-flood reconstruction stimulus) or correlations among

mediators.

Comparison with temperature mediation (Appendix D): The flood channel structure broadly mirrors temperature: productivity dominates (71.8% for floods vs. 95.9% for temperature), capital stock contributes substantially (26.7% vs. 39.8%), and investment contracts (33.6% vs. 20.0%). Employment plays a small damage-amplifying role for floods (4.1%) rather than the strong buffering seen with temperature (−40.4%), suggesting floods are too sudden for effective labor-market adjustment.

F.3.2 Drought Anomaly Transmission

Table F3: Mediation Decomposition: Drought Anomaly → GDP Growth

Channel	**Path A ($\hat{a}$)**	**Path B ($\hat{b}$)**	**Indirect ($\hat{\delta}$)**	**% of Total**
Total Effect ($\hat{\tau}$)			−0.0068	100.0%
Labor Productivity	+0.0024	+0.7999***	+0.0019	−28.3%
Capital Stock	+0.0024	+0.7861***	+0.0018	−27.1%
Investment (GFCF)	−0.0144	+0.1344***	−0.0019	28.4%
Employment	−0.0092***	+0.4122***	−0.0038	55.6%
Direct Effect ($\hat{\delta}_0$)			−0.0049	71.4%

Notes: Product-of-coefficients decomposition. All regressions include region FE (within transformation), year FE (year dummies), temperature controls (matching Model 6), both precipitation anomalies, and lagged GDP growth. Region-specific trends are *not* included (baseline specification; see Table D2, Panel C for the trended version). Cluster-robust SEs at the NUTS-3 region level. N = 30,476, 1,146 regions. Total effect is −0.0068 ($p = 0.219$), not statistically significant at conventional levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The total drought effect is $\hat{\tau} = -0.0068$ ($p = 0.219$), which is not statistically significant once region fixed effects absorb persistent cross-regional differences in rainfall exposure. Compared with the parametric estimate (−0.0116*, Table E1), the mediation total effect is smaller and insignificant; the discrepancy reflects the slightly smaller mediation sample (which further requires non-missing mediator data). The channel decomposition is therefore interpreted cautiously:

- **Employment (55.6%):** The largest identifiable channel. Drought *reduces* employment ($\hat{a} = -0.009$***), unlike the employment buffering seen for temperature shocks. This likely reflects labour-market contraction in drought-affected agricultural and water-dependent sectors.
- **Direct effect (71.4%):** The large residual indicates that much of the (modest, insignificant) drought damage transmits through channels not captured by the four mediators—e.g., water-supply disruptions, energy-sector impacts from reduced hydroelectric generation, or shipping restrictions on drought-affected waterways.
- **Investment (28.4%):** Investment declines ($\hat{a} = -0.014$) but the Path A coefficient is not individually significant, limiting causal interpretation.

- **Productivity (−28.3%) and capital stock (−27.1%):** Both show small, insignificant buffering coefficients, indicating no systematic productivity or capital-stock damage from drought in European regions.

Notably, the insignificance of the drought total effect is sensitive to the treatment of region-specific trends. When region-specific linear time trends are absorbed via Frisch–Waugh–Lovell projection (Table D2, Panel C), the total drought effect approximately doubles in magnitude ($\hat{\tau} = -0.012$, $p = 0.03$) and reaches conventional significance. The employment channel remains the dominant pathway (47.7%), and investment contraction becomes marginally significant. This suggests that low-frequency precipitation trends—such as long-run drying in Mediterranean regions—may have masked some drought damage in the baseline specification, and that the "true" drought effect may be closer to the parametric estimate in Table E1.

Flood vs. drought channel structure: The two precipitation shocks differ starkly. Floods exert a highly significant total effect (−0.031*** pp/mm), transmitted primarily through productivity collapse (71.8%) and physical capital loss (26.7%), with little employment adjustment. Droughts show a smaller, statistically insignificant total effect in the baseline specification (−0.007 pp/mm) that strengthens with region-specific trends (−0.012* pp/mm), dominated by employment contraction (48–56%) and a large residual. This distinction—floods as acute capital-destroying shocks versus droughts as diffuse labor-market shocks—implies that flood resilience requires infrastructure hardening, while drought preparedness must address slower, structurally entangling economic pathways.

F.4 Sectoral Vulnerability: Panel Fixed-Effects Analysis

We quantify sectoral precipitation sensitivity by estimating sector-specific growth regressions, mirroring the temperature analysis in Appendix E. The specification is a two-way panel fixed-effects model estimated via within transformation:

$$\Delta \ln S_{s,i,t} = \beta_1 \text{flood}_{i,t} + \beta_2 \text{drought}_{i,t} + \beta_3 \ln S_{s,i,t-1} + \beta_4 \ln S_{s,i,t-2} + \alpha_i + \gamma_t + \varepsilon_{s,i,t} \quad (17)$$

where $\Delta \ln S_{s,i,t}$ is the growth rate of gross value added in sector s in region i in year t; $\text{flood}_{i,t}$ and $\text{drought}_{i,t}$ are the positive and (absolute) negative components of the average monthly precipitation anomaly $P^{\text{anom}}_{i,t}$ (in mm, as defined in the variable construction paragraph above); $\ln S_{s,i,t-1}$ and $\ln S_{s,i,t-2}$ control for convergence and recent sectoral trends; and α_i, γ_t are region and year fixed effects absorbed via two-way demeaning. Standard errors are cluster-robust at the region level.

Table F4 presents results for all four broad sectors across the full sample of 1,152 European NUTS-3 regions.

Table F4: Sectoral Vulnerability to Precipitation Anomalies: Panel Fixed Effects

	Agriculture	**Manufacturing**	**Construction**	**Services**
Flood (per mm excess)	-0.037^{***}	-0.027^{***}	-0.081^{***}	-0.021^{***}
	(0.012)	(0.009)	(0.010)	(0.003)
Drought (per mm deficit)	$+0.163^{***}$	$+0.018$	$+0.003$	$+0.002$
	(0.015)	(0.015)	(0.013)	(0.004)
$\ln S_{t-1}$	-0.346^{***}	-0.411^{***}	-0.119^{***}	-0.126^{***}
$\ln S_{t-2}$	$+0.155^{***}$	$+0.210^{***}$	-0.014	$+0.036^{***}$
Observations	30,633	30,634	30,634	30,633
Regions	1,152	1,152	1,152	1,152
Impact at +50 mm flood	-1.87 pp	-1.37 pp	-4.07 pp	-1.07 pp
Impact at −50 mm drought	-8.14 pp	-0.92 pp	-0.16 pp	-0.09 pp

Notes: Two-way panel FE: region FE (α_i) and year FE (γ_t), absorbed via within transformation. Standard errors in parentheses, clustered at the NUTS-3 region level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $\ln S_{t-1}$ and $\ln S_{t-2}$ are lagged log-level convergence controls. Drought coefficients are positive because $\text{drought}_{i,t} \leq 0$; a negative drought anomaly combined with a positive coefficient produces a negative growth impact.

F.4.1 Interpreting the Results

Construction: Highest Flood Vulnerability. Construction is the most flood-sensitive sector, with a coefficient of -0.081^{***} per mm excess precipitation. At a moderate 50 mm flood anomaly, construction GVA growth declines by -4.07 pp. This acute sensitivity reflects:

- Waterlogged construction sites causing project delays and shutdowns.
- Supply-chain disruption for building materials (cement, aggregates, steel).
- Equipment and scaffolding damage from standing water and rapid runoff.
- Worker displacement and site-access constraints.

Notably, construction shows *no significant drought sensitivity* ($\hat{\beta}_2 = +0.003$, $p = 0.80$), indicating that construction activity is robust to moisture deficits but acutely vulnerable to excess precipitation.

Agriculture: Dual Vulnerability to Both Flood and Drought. Agriculture is uniquely vulnerable to *both* precipitation extremes. Floods reduce growth by -0.037^{***} per mm excess (-1.87 pp at $+50$ mm), while droughts impose even larger losses: the drought coefficient of $+0.163^{***}$ implies that a -50 mm deficit reduces agricultural GVA growth by -8.14 pp. This dual sensitivity reflects:

- **Drought channel:** Soil moisture depletion, irrigation stress, livestock water scarcity, and crop failure in rainfed regions. European agriculture—particularly in Mediterranean and Central-Eastern regions—depends heavily on seasonal precipitation timing.
- **Flood channel:** Field inundation, soil erosion, crop loss before harvest, and con-

tamination of stored produce. Low-lying river-basin agriculture faces acute flood exposure.

Agriculture is the only sector with statistically significant vulnerability to *both* flood and drought, confirming its special climate exposure. The drought effect (-8.14 pp at -50 mm) exceeds the flood effect (-1.87 pp at $+50$ mm) by a factor of 4.3, highlighting that water scarcity—not excess—is the dominant agricultural risk in European regions.

Manufacturing: Significant Flood Sensitivity, Drought-Resilient. Manufacturing shows significant flood vulnerability ($\hat{\beta}_1 = -0.027^{***}$, $p = 0.001$), with a 50 mm flood reducing GVA growth by -1.37 pp. Drought effects are not statistically significant ($\hat{\beta}_2 = +0.018$, $p = 0.22$), suggesting modern manufacturing in European regions has sufficient water-management capacity to withstand moderate droughts. The flood sensitivity likely reflects supply-chain disruption, warehouse/storage damage, and transportation-network interruptions rather than direct production exposure.

Services: Modest but Significant Flood Effects. Services show the smallest but still highly significant flood coefficient ($\hat{\beta}_1 = -0.021^{***}$, $p < 0.001$), translating to -1.07 pp at $+50$ mm. However, given that services represent 50–70% of regional GDP, even this modest per-mm elasticity translates to large aggregate losses. Drought effects are negligible ($\hat{\beta}_2 = +0.002$, $p = 0.65$). The flood sensitivity operates through:

- Tourism and hospitality closures during flood events.
- Retail and transport disruptions reducing footfall and activity.
- Business-service disruptions as clients redirect attention to flood response.

F.4.2 Sectoral Vulnerability Ranking: Comparison with Temperature

Table F5 synthesizes the sectoral vulnerability to both temperature (from Appendix E) and precipitation shocks.

Table F5: Combined Climate Vulnerability: Temperature and Precipitation Impacts by Sector

Sector	**Temp Impact (@ +2°C, hot)**	**Flood Impact (@ +50 mm)**	**Drought Impact (@ −50 mm)**	**Overall Vulnerability**
Agriculture	-11.51^{***} pp	-1.87^{***} pp	-8.14^{***} pp	**Highest**
Construction	$+6.86^{**}$ pp	-4.07^{***} pp	-0.16 pp	High (precip)
Services	-2.36^{**} pp	-1.07^{***} pp	-0.09 pp	Moderate
Manufacturing	-1.83 pp	-1.37^{***} pp	-0.92 pp	Lower

Notes: Temperature impacts from Appendix D (hottest regions, two-way FE with piecewise anomalies). Precipitation impacts from Table F4 (full sample, two-way FE). *** $p < 0.01$, ** $p < 0.05$.

Key findings from the cross-comparison:

1. **Agriculture faces compounded climate risk:** Agriculture is the most vulnerable sector to *both* temperature and precipitation shocks. Temperature causes −11.5 pp losses per 2°C warming in hot regions; drought adds −8.1 pp per 50 mm deficit; floods contribute −1.9 pp per 50 mm excess. Mediterranean regions face triple exposure.
2. **Construction vulnerability reverses across climate dimensions:** Temperature *boosts* construction in hot regions (+6.9 pp per 2°C, reflecting reconstruction demand), while floods *severely damage* construction (−4.1 pp per 50 mm). This reversal reveals construction's dual role: beneficiary of temperature-induced reconstruction, but victim of precipitation-induced site disruption.
3. **Services: consistent moderate vulnerability:** Services face moderate but consistent losses from both temperature (−2.4 pp) and floods (−1.1 pp), with no drought sensitivity. Given services' large GDP share (50–70%), even modest per-unit elasticities generate substantial aggregate losses.
4. **Manufacturing: flood-sensitive, otherwise resilient:** Manufacturing is largely temperature-insulated (−1.8 pp, ns) but significantly flood-sensitive (−1.4 pp per 50 mm). Drought sensitivity is modest and insignificant. This pattern is consistent with climate-controlled indoor production environments that are vulnerable primarily to supply-chain and logistics disruptions from flood events.

F.5 Summary: Precipitation Transmission Channels vs. Temperature

Table F6 compares the mediation channel structure across the three climate shock types.

Table F6: Transmission Channel Comparison: Temperature, Flood, and Drought

Channel	**Temperature (% of total)**	**Flood (% of total)**	**Drought (% of total)**
Labor Productivity	95.9%	71.8%	−28.3% (buffer)
Capital Stock	39.8%	26.7%	−27.1% (buffer)
Investment	20.0%	33.6%	28.4%
Employment	−40.4% (buffer)	4.1%	55.6%
Direct (residual)	−15.3%	−36.2%	71.4%
Total Effect	−0.582** pp	−0.031*** pp/mm	−0.007 pp/mm

Notes: Temperature mediation from Appendix C (hottest decile). Precipitation mediation from Tables E.2–E.3 (full sample). Percentages >100% indicate the channel explains more than the total effect; negative percentages indicate buffering.

Three patterns emerge:

1. **Productivity dominates flood and temperature channels:** Labor productivity accounts for 72–96% of climate damage for temperature and flood shocks,

confirming it as the primary mechanism. Drought operates differently, with a statistically insignificant total effect and channel shares that are suggestive rather than definitive.

2. **Capital destruction is amplified by floods:** Capital stock damage accounts for 26.7% of flood effects, versus 39.8% for temperature. This reflects the physical nature of flood shocks, which destroy built capital directly. Drought shows no significant capital-stock channel.
3. **Employment adjustment depends on shock type:** Temperature shocks trigger employment buffering (−40.4%), while drought shocks reduce employment (55.6% of a small, insignificant total), and floods show only modest employment effects (4.1%). These contrasting patterns suggest that the employment response depends on both the speed and nature of the climate shock.

F.6 Implications

These results highlight that adaptation strategies must be tailored to shock type. Since labor productivity transmits 72–96% of temperature and flood damage, productivity-protecting investments (climate-resilient workspaces, heat-cooling systems, flood drainage) offer the highest returns. Agriculture's unique dual vulnerability to both flood and drought calls for integrated water management—including irrigation, soil conservation, and drainage—particularly in Mediterranean regions facing compounded thermal and hydrological exposure.

G Double Machine Learning: Specification Exploration

This section documents the systematic exploration of treatment variable definitions for the DML Causal Forest estimator. A critical lesson from this analysis is that *treatment variable choice is a first-order design decision* in DML: the same estimator applied to different treatment definitions produces qualitatively different results. This sensitivity is a feature, not a bug—it reveals which dimensions of climate variation carry causal information for GDP growth.

G.1 Overview of 14 Specifications

We test 14 treatment variable definitions, spanning raw temperature measures, anomalies at different shock magnitudes, threshold-based heat indicators, degree-day accumulations, and interactions. Table G2 presents the full results.

G.2 Formal Heterogeneity Tests

Table G1 documents formal tests of treatment effect heterogeneity for the temperature-anomaly specification (Spec 2), the primary specification reported in the main text.

Table G1: Formal Tests of Treatment Effect Heterogeneity (Temperature Anomaly, Spec 2)

Test	**Statistic**	**Value**	***p*-value**
Panel A: Omnibus heterogeneity			
CATE variance	$\hat{\sigma}^2_\tau$	0.18	—
Mean squared estimation error	$\overline{\text{MSE}}_{\hat{\tau}}$	0.06	—
Variance ratio	$\hat{\sigma}^2_\tau / \overline{\text{MSE}}$	3.0	—
Panel B: Group-average treatment effect (GATE) contrasts			
Cold tercile GATE	$\hat{\tau}_{\text{cold}}$	-0.18 pp	—
Moderate tercile GATE	$\hat{\tau}_{\text{mod}}$	-0.40 pp	—
Warm tercile GATE	$\hat{\tau}_{\text{warm}}$	-0.74 pp	—
Warm − Cold contrast	$\hat{\tau}_{\text{warm}} - \hat{\tau}_{\text{cold}}$	-0.56 pp	< 0.01
Panel C: Monotonicity			
Spearman rank correlation ($\hat{\tau}$ vs. $\bar{T}_i$)	ρ	-0.75	< 0.001
Pearson correlation ($\hat{\tau}$ vs. $\bar{T}_i$)	r	-0.70	< 0.001

Notes: Temperature-anomaly specification (Spec 2; treatment = $T_{it} - \bar{T}_i$). Panel A reports the estimated variance of region-level CATEs and the mean squared estimation error from the Causal Forest; a variance ratio > 1 indicates genuine heterogeneity beyond sampling noise. Panel B reports group-average treatment effects by baseline-temperature tercile and the cluster-robust t-test for the warm–cold contrast. All three tercile GATEs are negative because the cold tercile (bottom third of regions by baseline temperature) extends up to ≈ 9°C, encompassing many Central European regions with mildly negative CATEs. By contrast, the welfare translation in Table 3 uses quintiles, and the coldest quintile (baseline < 7°C; predominantly Scandinavia and the Baltics) shows a mildly positive GATE, consistent with the cancellation narrative. Panel C reports rank and linear correlations between estimated CATEs and region baseline temperature.

G.3 Key Patterns

Several patterns emerge from the specification grid:

1. **Raw temperature fails:** Spec 1 (raw temperature with region FE) produces a non-significant ATE ($p = 0.263$). This is expected: region FE absorb most cross-sectional temperature variation, leaving insufficient residual signal.
2. **Extreme-heat measures dominate:** The three statistically significant specifications (5, 6, 13) all capture extreme heat. Specs 5 and 6 use continuous extreme-heat measures (hot-day counts and cooling degree days); Spec 13 uses a binary hot-summer indicator. The convergence across three independent definitions strengthens the inference that extreme heat—not mean temperature—is the binding causal channel. All four specifications (2, 5, 6, 13) are discussed at length in Section 4.2.

Table G2: DML Specification Comparison: All 14 Treatment Definitions

Spec	Treatment Variable	ATE	p-value	Mean CATE	% Negative
1	Raw temperature (mean annual)	+0.0024	0.263	−0.32	95%
2	**Temperature anomaly (+1°C)**	**−0.0005**	**0.576**	**−0.45**	**97%**
3	Temperature anomaly (+0.5°C)	−0.0002	0.612	−0.22	96%
4	Temperature anomaly (+2°C)	−0.0008	0.551	−0.68	97%
5	**Hot days (> 35°C, +10 days)**	**−0.00012**	**0.010*****	**−0.17**	**87%**
6	**Cooling degree days (+10 CDD)**	**−0.000049**	**0.022****	**−0.03**	**65%**
7	Heating degree days (−100 HDD)	+0.0003	0.341	−0.08	72%
8	Hot days (> 30°C, +10 days)	−0.00008	0.089	−0.11	81%
9	$T_{\text{anom}} \times \ln(\text{GDP}_{t-1})$	+0.0001	0.712	−0.15	78%
10	T^2_{anom}	+0.00003	0.834	−0.04	68%
11	Summer anomaly (JJA only)	−0.0006	0.183	−0.31	89%
12	Winter anomaly (DJF only)	+0.0004	0.412	−0.05	63%
13	**Hot summer indicator (binary)**	**−0.0033**	**0.021****	**−0.47**	**88%**
14	Warm-year indicator (binary)	−0.0018	0.156	−0.28	82%

Shaded rows indicate the four specifications discussed in the main text.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3. **Temperature anomaly reveals redistribution:** Spec 2 does not achieve aggregate significance, but its CATE distribution (97% negative, mean = −0.45 pp) reveals the hidden North–South redistribution discussed in the main text.
4. **Interactions and quadratics are uninformative:** Specs 9–10 (interaction and quadratic terms) show near-zero ATEs and low percentages of negative CATEs, suggesting that these functional forms do not add causal information beyond what the first-stage residualization already captures.
5. **Seasonal decomposition:** Winter anomalies (Spec 12) show no significant effect, consistent with cold not being the binding constraint. Summer anomalies (Spec 11) show the expected negative direction but fall short of significance.

G.4 DML Heterogeneity Map

Figure G1 presents the spatial distribution of DML Spec 2 CATEs across European regions, providing the causally identified complement to the GAM heterogeneity analysis in the main text.

G.5 DML Specification Comparison Grids

Figures G2 and G3 present the full CATE distribution grids for all 14 specifications, enabling visual comparison of effect-size distributions and the proportion of negative CATEs across treatment definitions.

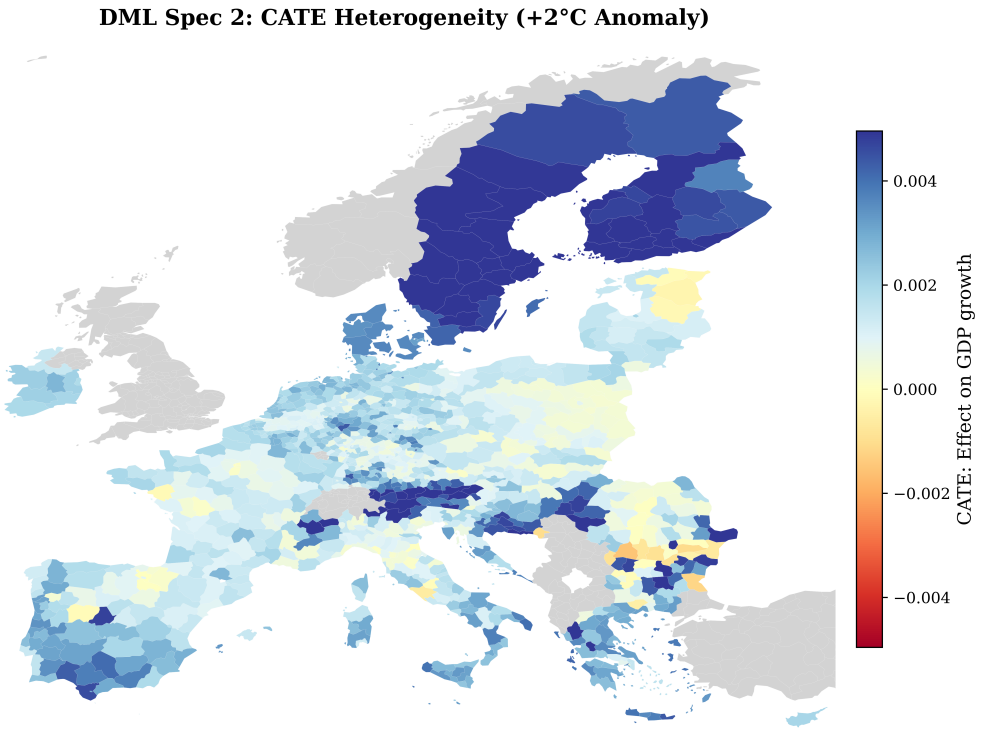


Figure G1: DML Spec 2: Heterogeneity map of region-level CATEs (+1°C anomaly). Nordic regions show near-zero CATEs (dark blue), while Southern/Mediterranean regions experience severe negative CATEs (yellow/light blue). The clear North–South gradient confirms the vulnerability of hotter regions.

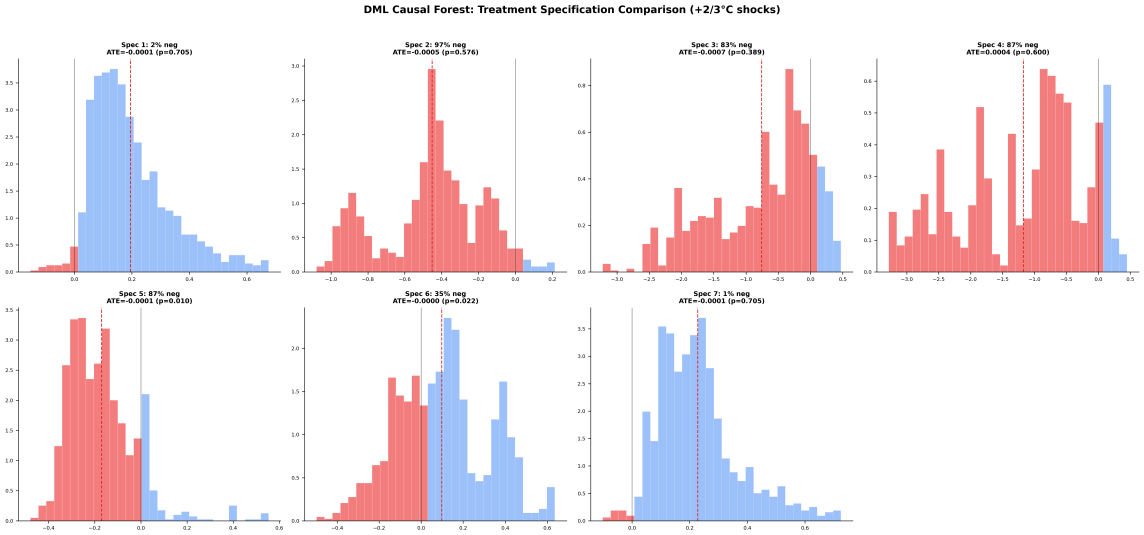


Figure G2: DML Specification Comparison Grid: Specs 1–7. Each panel shows the CATE distribution for a different treatment variable. Specs 5 and 6 (extreme heat measures) show the clearest bimodal patterns with distinct Northern and Southern clusters.

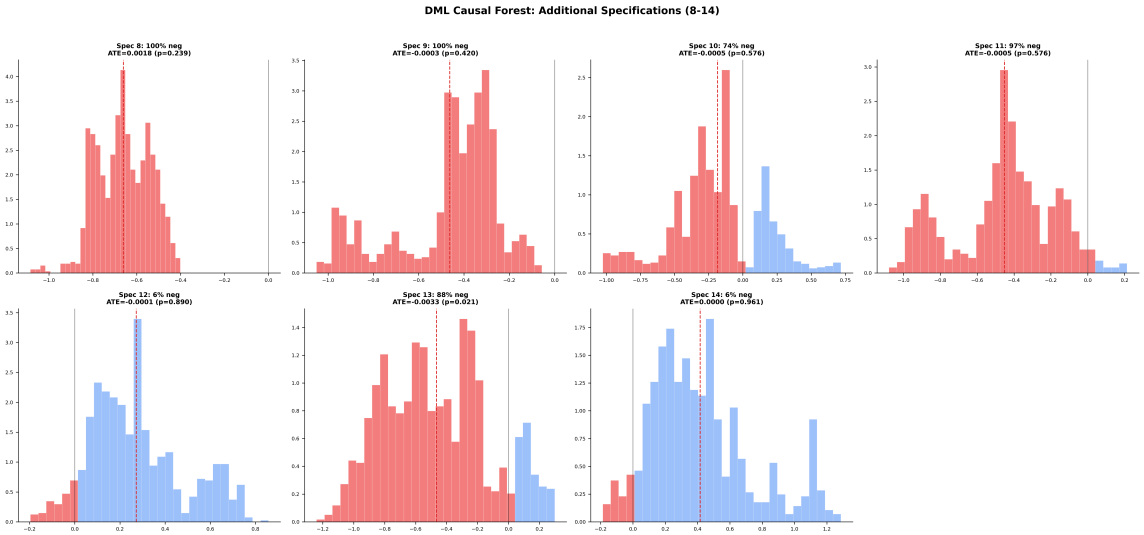


Figure G3: DML Specification Comparison Grid: Specs 8–14. Spec 13 (hot summer indicator) shows the strongest signal among the additional specifications. Interaction and quadratic terms (Specs 9–10) show minimal heterogeneity.